

Universe24: the group of order 24 behind the clicks, one local integer operator on the cube's lattice, and the quantum and relativistic experiments it confronts

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Abstract

Can one local integer operator on a cubic lattice, read only through detectors' clicks, confront the quantum and relativistic experiments? The whole GameBoard is one element of one integer module, and the law is one linear operator on it: at every Node a second-order step on the Node's own two levels and its six neighbours, with one declared integer pair per family and the remainder of the one division kept, and at declared cells a coupling between two families and an integer label matrix. The lattice's symmetries are the 48 signed permutations of the axes and their 24 rotations. Two acts complete the law. The click is the only non-local act: when a record's one-way inward flux into a detector, a cube of side at least 3, crosses the record's threshold, the record ends whole, everywhere, at once. The birth is the only write, the other side of a click. Exact on the lattice: the conserved quadratic form of every family and of two coupled families, the step's bijection, the local flux identity, one click per quantum, Born's rule as a detector's share of a record's inward flux to the grain $1/W$ of the record's wheel, and no signalling through the click. A product state gives the Clauser–Horne–Shimony–Holt (CHSH) sum at most 2; an entangled pair read on integer polariser axes gives $14/5$. Recovered as the wave number goes to 0: $c = 1/\sqrt{3}$ as the band's pace, isotropic at second order by the 48; and a moving packet's clock slows by its own dispersion, with no rule for it. An experiment is a list of entries on the board; its initial state is generated once by the operator itself and checked in integers at load. Seventeen experiments are closed in the algebra with pins written blind; none is run yet on this engine.

Keywords: integer lattice model; the symmetry group of the cube; one linear operator; entangled pair; Bell correlations; discrete relativity; pre-registered simulation

1 Introduction

The group was found, not assumed. The operations came first, as acts on the board, and the road is told in Section 7: a message crossed one Link per interval, messages met at a Node, a detector clicked. The group is the answer to which maps preserve what the acts work on, a Node with six Ports and its Links and a causal cone of one Link per interval: the bijections of the six Ports that keep opposite Ports opposite are the 48 signed permutations of the axes, and the only $\mathbb{Z}$ -linear bijections that preserve the cone are these 48, so no boost is among them and Lorentz's form is reached Outside from the clicks, not as a symmetry of the lattice. The 24 came from the hand, a pseudoscalar that a reflection flips and a rotation keeps, which splits the 48 into the 24

rotations and the 24 reflections. Every rule is required to commute with the 48, a requirement beyond locality. Today the law is one operator on one element, with no stepping body and no merge of messages [9, 9.19].

The whole move, in the order it is built. Seven things, each named once in the project's terminology [32].

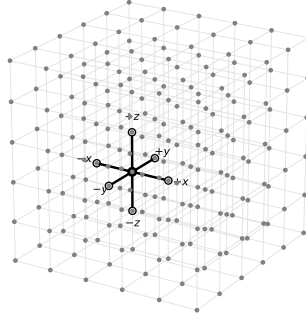
- (1) *The GameBoard and its group.* The board is a cubic lattice of Nodes, six neighbours each. The symmetry group of the cube is the group of order 48: the 24 rotations, the group of order 24, and the reflections. The law is the same at every Node and commutes with the 48 and with the translations, and two things follow: the pace of light is the same in every direction at leading order, $c^2 = 1/3$ from the six reads, and the departure from it appears only in a short wave, in the fixed cubic pattern of the 48 (Eq. (12)).
- (2) *The one element.* The state of the whole board, every Node, every family and every label together, is one element $\mathbf{v}$ of one integer module: one summand per quantum, a *record*, with its two levels (now and one interval before) at every Node and label and the remainder of the one division [9, 9.19 (1)]. A *family* is an index of the summands: its vacuum pair [num, den] and its label module, and nothing else [9, 9.22 (1)]. A world has at most three: light, the experiment's matter and the holder of the tools' bodies.
- (3) *The one operator.* The law is one linear map $\mathbf{S}$ of that element (Eq. (1)): per family the second-order step with the family's pair at every Node, and at declared cells a coupling between two families and an integer label matrix [9, 9.19 (2)].
- (4) *The tools.* A tool is a region of the operator: a cube of the holder family with its material integers (a pair, a coupling, an axis), its total momentum $\mathbf{K}$ (the wave vector of its character, $\mathbf{0}$ at rest), at most a birth, and a name [9, 9.19 (5)]; no code belongs to a tool [10, A].
- (5) *The detector and the click.* A detector is a named region, a cube of side at least 3. The click is the only non-local act: a record's one-way inward flux into the detectors named on its ladder, summed over intervals, crosses the record's threshold, and the record ends whole, everywhere, at that interval; the click is the detector's and names no Node inside it [9, 9.25].
- (6) *The birth.* The only write, the other side of a click: the click of an emitter body's own excited record writes the born record's two levels once on the emitter's cells, the transpose $\mathbf{E}^T$ of the reading, and from then on only the law advances it [9, 9.13, 9.17].
- (7) *The experiment.* A list of entries on the board, each its family, how much and where, with its momentum; a generator writes the initial state once from the operator itself, the engine checks it in integers at load and runs the law, and the output is the clicks, compared with a pin written before the run (Section 2, "The experiment as data"; Section 6).

The general formula. The law is one step $\mathbf{S}$ of the one element at every interval, and two acts on its summands:

the step: $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M}_F \mathbf{a}_{\text{now}} + \mathbf{c}_F$, $\mathbf{M}_F = \mathbf{D}_F^{-1} \mathbf{A}/3$, per family F, with the remainder kept; the click: $2W C(t^*) \geq (2u + 1) T$, $C(t) = \sum_{s \leq t} \sum_{\text{Ports } j \rightarrow i \text{ into the ladder's detectors}} \max(G_{ij}(s), 0)$; the birth: $(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}}) \leftarrow \mathbf{E}^T(\text{the born pair})$ on the emitter's cells, at the click's interval.

(1)

In the step, $\mathbf{a}_{\text{now}}$, $\mathbf{a}_{\text{before}}$ and $\mathbf{a}_{\text{next}}$ are a record's levels over the Nodes now, one interval before and one interval on; $\mathbf{A}$ is the read matrix of the board (the six reads of every Node); $\mathbf{D}_F$ is the diagonal matrix of the per-Node weights $\text{den}_i/\text{num}_i$ of the family's pair; and $\mathbf{c}_F$ is the coupling's first difference of the other family's levels at the cells of a coupled body, with, at the cells of a polar



one Node and its six Ports, the Links to its six neighbours

Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Figure 2. Each record keeps at a Node its two levels and its remainder there, and a body's region keeps its material integers; nothing else sits at a Node. Drawn from the definitions; no run.

body, the integer label matrix on the record's two label rows. In integers the step is Eq. (2), one division by 3 den_i with the remainder kept. In the click, G_{ij} is the flux into Node i from Node j (Eq. (5)), T the record's conserved form, u its residue and W its wheel, both read from the law at its birth (Definition 1); the click falls at the first interval t^* at which the inequality holds, at the detector whose share of that interval's flux holds the threshold, and deletes the record. The birth is fired only by a click. Nothing else happens on the board: no table, no rate, no drive, no take and no declared residue. Section 2 states each piece with its proof or its mark; every derivation of the paper names the piece it starts from.

Physics as the passage of information. What the law describes is the passage of information from an emitter to a detector, and nothing between them is read. A record is born at an emitter's click with its two levels written once; it moves on the lattice by the one operator, at most one Link per interval; it leaves at a click, where the record ends whole, and the click is what is read: reality Outside is the clicks, and a number of the board is a diagnostic. Transport is local at every step; the one non-local act is the click itself, the record ending everywhere at once, and no information passes through it (Theorem 4). The road by which this object was found is Section 7; the engine on which it is computed is Section 6.

The reading rule and the notation. Only a detector's click is a measurement, and an experiment is compared with nature and with its pin through its receivers' clicks alone; the operator, the tools, conservation and the initial state are proved in the algebra and checked by the property test and the load check, never by an experiment's run [9, 9.21 (4)]. Every physics name in this paper (Born's, Tsirelson's, Malus's, Lorentz's) names the known form that the law's result recovers. A number of the GameBoard's state is a diagnostic of the host, labelled so, never compared with nature and never pinned. Every symbol is named in English at its first use, a scalar plain, a vector in bold lowercase, a matrix or an operator in bold uppercase.

What this paper contributes, and what it does not claim. New here: one explicitly stated integer operator with its click and its birth, whose conserved forms, bijection, flux identity, Born's rule and no-signalling are theorems on the lattice; experiments as short lists of entries, their initial states generated from the operator itself and checked in integers at load; and seventeen experiments

closed in the algebra with pins written blind before any run. Not claimed: a derivation of the Lorentz group, which is not among the 48; a moving rigid rod, which the model does not yet define (it has clocks and no rods, [9, 9.24 (6)]); or any comparison with nature before the experiments are run on this engine. Gravity is not in this paper.

2 The algebra: one element, one operator, the click and the birth

This section states the object and nothing else: the board and its group, the one element, the one operator with its conserved forms, the flux, the click, the birth, the families and the experiment as data. Each statement carries the mark of the algebra document [9]: PROVED (a proof for every board, in the document or here), DERIVED (from proved statements, by argument), COMPUTED (a computation, not a proof), or CARRIED (a design's proof, cited).

The definitions, in order. Five objects, each one line of mathematics before its words.

- (1) The GameBoard: the Nodes of a box in $\mathbb{Z}^3$ whose axes are periodic or open; the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$, with the involution $e \mapsto -e$ (the opposite Port); at most one Link per interval.
- (2) The group of order 48 and its rotations: $G_{48} = \{g : P \rightarrow P \text{ a bijection with } g(-e) = -g(e)\}$, the 48 signed permutations of the three axes; $\det : G_{48} \rightarrow \{+1, -1\}$ a homomorphism; $G_{24} = \ker \det$, the rotations, the group of order 24 (Theorem 1). The hand, a pseudoscalar, $h \mapsto \det(g) h$, tells the two cosets apart.
- (3) The one element: $V = \bigoplus_q V_q$, $V_q = (\mathbb{Z}^{\text{Nodes}} \otimes L_q)^2 \otimes (\text{the remainders})$, one summand per quantum q , L_q its label module ($\mathbb{Z}^2$ per arm; $\mathbb{Z}^2 \otimes \mathbb{Z}^2$ for a record of rank 2); the state of the world at an interval is one element $\mathbf{v} \in V$ [9, 9.19 (1)].
- (4) The one operator: $\mathbf{S} : V \rightarrow V$, per summand of family F the step of Eq. (1), plus the coupling between summands of coupled families and the label matrix on the axes' cells; linear, local and equivariant [9, 9.19 (2), (6)].
- (5) The two acts: the click, a record's one-way inward flux $\mathbf{E}$ against its form, the rung and the deletion of the summand, the only non-local act; and the birth $\mathbf{E}^\top$, the only write, at the click's interval (Definitions 1 and 2).

The words of this paper. The *GameBoard* is the cubic array of Nodes, each joined to six neighbours by Links through directional Ports; time is a sequence of intervals. *Inside* is the board and what moves on it, where no one measures; *Outside* is the game above the board, the detectors and their clicks, the only place a number is read. A *record* is one summand of the element, one quantum: its two levels over the Nodes and its labels, with the remainder kept per Node on the record's own row. A *family* is an index of the summands, its vacuum pair and its label module. A *body* is a region of the operator where a family's pair differs from the vacuum's; a *tool* is a region with its integers, at most a birth and a name; a *face* is the operator's zero row beyond an open border; a *receiver* or *detector* is a name the host gives a region, on records' ladders [9, 9.19 (5)]. Every number carries its *kind*: DETECTOR, a click or a count of clicks, the only measurement; GAMEBOARD, a number of the board's state, a diagnostic and never a result; COMPUTATION, a number computed from the algebra before any run; DECLARATION, a declared integer of an experiment; HOST, a number of the host's bookkeeping, never a reading. Figure 1 draws the board in space.

The six Ports, the group and the front. Everything geometric in the law follows from one Node and its six Ports, $\pm x, \pm y, \pm z$, and from at most one Link per interval. The maps of the six

Ports onto themselves that keep opposite Ports opposite are the signed permutations of the three axes, 48 in all, the symmetry group of the cube and of the octahedron. The determinant splits them into two halves of 24: the rotations, which permute the cube's four body diagonals, and the improper maps, the reflections with the central inversion among them.

Theorem 1 (the 24 of the name). *The maps of the six Ports onto themselves that send opposite Ports to opposite Ports form the group of the signed permutations of the three axes, of order $2^3 \cdot 3! = 48$. The determinant is a homomorphism of this group onto $\{+1, -1\}$; its kernel, the group of order 24 (the rotation group of the cube), has index 2 and is isomorphic to the symmetric group S_4 of the permutations of the cube's four body diagonals; the other coset, the reflections, has 24 elements too. The hand, a pseudoscalar that a map g sends to $\det(g)$ times itself, tells the 24 rotations from the 24 reflections, as does an axial vector.*

Proof. A map that keeps opposite Ports opposite is fixed by the images of $+x, +y, +z$, which must be three Ports no two of them equal or opposite: $6 \cdot 4 \cdot 2 = 48$ choices, each a signed permutation matrix, and every signed permutation matrix keeps opposite Ports opposite. The determinant of a product is the product of the determinants, so $\det$ is a homomorphism, onto since -1 has determinant -1 ; its kernel has index 2, order 24. A rotation permutes the four lines through opposite vertices of the cube, and only the identity fixes all four, so the kernel embeds in S_4 , of order 24: the two are isomorphic. $\square$

The rule per family. At every Node i a record's row $(a_{\text{now}}, a_{\text{before}}, r)$ is mapped once per interval by

$$3 \text{den}_i a_{\text{next}} + r' = \text{num}_i S_6 - 3 \text{den}_i a_{\text{before}} + r, \quad 0 \leq r' < 3 \text{den}_i, \quad (2)$$

then $a_{\text{before}} \leftarrow a_{\text{now}}, a_{\text{now}} \leftarrow a_{\text{next}}, r \leftarrow r' [9, 8.1]$. Here S_6 is the sum of the six reads of the record's a_{now} (the neighbour across a Link; the Node itself on a folded axis of extent 1; 0 beyond an open face), $[\text{num}_i, \text{den}_i]$ the family's pair at the Node (the vacuum's pair everywhere, a body's pair on its cells) and r the remainder of the division, kept on the record's row. Light is the pair $[1, 1]$; a massive family is a pair with $\text{den} > \text{num}$, and nothing else names the mass. The step is made of three verbs on the row: the bilinear form (B), one entry num_i on the six reads and one entry -3den_i on the row's past; the translation (T) of the row by that rate; and the division (D) by the wall 3den_i with the remainder kept. The three tests hold [9, 8.1]: generic, one primitive with one declared pair and no family name; vector, the three verbs with no root and no float; local, the record's own row and its six reads, nothing kept at a Node beyond the row and its remainder.

The conserved form. Let $\mathbf{A}$ be the read matrix of the board (A_{ij} the number of reads of j among the six reads of i), symmetric with row sums at most 6, $\mathbf{D} = \text{diag}(D_i)$ with $D_i = \text{den}_i / \text{num}_i$, and $\mathbf{M} = \mathbf{D}^{-1} \mathbf{A} / 3$. Without the remainder the rule is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$, and the form

$$I(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = \sum_i D_i (a_{\text{next},i}^2 + a_{\text{now},i}^2) - \frac{1}{3} \sum_i a_{\text{next},i} (\mathbf{A} \mathbf{a}_{\text{now}})_i \quad (3)$$

is conserved on every board and every extent, since $\mathbf{DM} = \mathbf{A} / 3$ is symmetric; it is positive definite for every massive pair and semidefinite at light's pair; and with the remainders kept the change per interval is exactly $I(t) - I(t-1) = \sum_i (a_{\text{next},i} - a_{\text{before},i})(r_i - r'_i) / (3 \text{num}_i)$, an integer identity with no tolerance (PROVED [9, 8.2]). A static level has $I = 0$.

The modes and the stability condition. Restricted to one family, $\mathbf{S}$ is a second-order linear recurrence with the symmetric operator $\mathbf{D}^{-1/2}(\mathbf{A}/3)\mathbf{D}^{-1/2}$, whose eigenvectors are the board's *modes* and whose eigenvalues are $2 \cos \omega$, the angle ω being the mode's clock, its frequency per

interval; on a torus there is one mode per Node. A mode is *bound* when $2\cos\omega$ lies above the vacuum band's top and *free* inside the band. The form I is positive definite, and every mode a rotation, exactly when the largest eigenvalue is below 2; by Perron–Frobenius that one condition bounds both ends of the spectrum. A pair with $\text{num} \leq \text{den}$ at every Node is sufficient, not necessary. The loader's check is the spectral one: the composed operator's largest eigenvalue below 2, the file refused otherwise (PROVED [9, 9.19 (2)]).

The coupling of two families. At a cell of a coupled body R the two records' rows are coupled by one entry of verb (B) over the other record's two levels, the first difference both ways: the massive row gains $g(a_{l,\text{now}} - a_{l,\text{before}})$, the receive, and light's row gains $-G(a_{m,\text{next}} - a_{m,\text{now}})$, the source, with g and G the two declared coupling pairs at the body's cells (the capital G apart from the flux G_{ij} below, which always carries two Node indices; CARRIED [9, 8.5]). The scheme is the Euler–Lagrange scheme of a two-point Lagrangian, so it is symplectic and has the exact invariant

$$J = I_m + \alpha I_l + g \sum_{i \in R} D_i (a_{m,\text{next}} - a_{m,\text{now}})_i (a_{l,\text{next}} - a_{l,\text{now}})_i, \quad \alpha = g D_{\text{in}}/G, \quad (4)$$

I_m and I_l the two families' forms, D_{in} the weight on the body's cells and α one number of the body: conserved exactly on every board, before the remainders (PROVED [9, 8.5]). The amplitude form of the coupling is withdrawn as tachyonic, its mass matrix at zero wave vector having a negative eigenvalue.

The label matrix. Every record carries its rows per label value of each arm. A polar body with the integer axis $\mathbf{u} = (a, b)$ applies at its cells the integer matrix $\mathbf{M}_{\mathbf{u}} = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ to the record's pair of label rows, verb (B), with no division: the common factor $a^2 + b^2$ scales the record's form by Brahmagupta's identity $(ax + by)^2 + (-bx + ay)^2 = (a^2 + b^2)(x^2 + y^2)$, exact on integers [9, 9.16 (2)]. The polariser's angle is this integer axis; there is no table of angles [9, 9.14 (b)].

The flux. Let $e_i = D_i (a_{\text{next},i}^2 + a_{\text{now},i}^2) - \frac{1}{3} a_{\text{next},i} (\mathbf{A}\mathbf{a}_{\text{now}})_i$ be the Node's share of the form, so that $I = \sum_i e_i$.

Proposition 1 (the local flux identity). *Between two intervals,*

$$e_i(t) - e_i(t-1) = \sum_{j \text{ read by } i} G_{ij}, \quad G_{ij} = \frac{1}{3} A_{ij} (a_{\text{now},i} a_{\text{before},j} - a_{\text{before},i} a_{\text{now},j}), \quad G_{ij} = -G_{ji}. \quad (5)$$

Proof. Substitute the rule $D_i (a_{\text{next},i} + a_{\text{before},i}) = (\mathbf{A}\mathbf{a}_{\text{now}})_i/3$ into the difference of e_i between the two intervals [9, 9.19 (3)]. $\square$

G_{ij} is the flux into Node i from Node j : a bilinear form of the record's own two levels at the two ends of a Link, free of the pairs (they cancel), antisymmetric and local. For a character moving from j to i it is positive. The one-way inward flux into one Node, summed over intervals and Ports of $\max(G_{ij}, 0)$, equals the record's form over the passage of a packet (COMPUTED to six digits on a chain [9, 9.19 (3)]). The rectification $\max(G, 0)$ is the click's own comparison: a record's energy is counted where it enters.

The residue and the wheel. Two things are called a remainder. The rule's remainder r is kept per Node on the record's row and is the law's own. A record's residue u is one integer, read once: the rule's remainder at the record's birth cell at the interval of the click that births it; its wheel

is $W = 3 \text{ den} / \text{gcd}(\text{num}, 3)$ of that cell's pair for the clicking record's family, in lowest terms. No residue, seed, order or wheel is declared anywhere; the loader refuses them (ADOPTED, record 1872 [6]; [9, 9.19 (4)]). From $r = 0$ every remainder at a Node is a multiple of $\text{gcd}(\text{num}, 3)$, so at most $3 \text{ den} / \text{gcd}(\text{num}, 3)$ values occur (PROVED); that they are reached and equidistributed is COMPUTED, not proved. The granularity of Born's rule on a wheel of W values biases a share by at most $1/(2W)$, so every birth cell carries a pair with $W \geq 500$, checked at load [9, 9.19 (4a)]. With the law's remainder the residues are equidistributed and not a permutation, and the counts are binomial about the shares, as nature's are; the run stays deterministic and reproducible.

Definition 1 (The detector and the click). A *detector* is one connected named region, a cube of side at least 3 (a smaller cube, or one name on disconnected pieces, is refused at load); separate places are separate names [9, 9.25 (7)]; record 1899 [6]. A record carries a *ladder*, the detectors it names in a declared order with the face receiver last. At its birth the record's threshold is fixed, $\theta = (2u + 1)T/(2W)$, with T the record's form I , and its running total starts at $C = 0$. At every interval t each detector d of the ladder receives the increment

$$f_d(t) = \sum_{\substack{i \in d, j \notin d \\ j \text{ read by } i}} \max(G_{ij}(t), 0), \quad (6)$$

the one-way inward flux through the region's Ports from outside it (a Link between two Nodes of one detector is no Port of it), and $C(t) = C(t-1) + \sum_d f_d(t)$. The *click* falls at the first interval t^* with $C(t^*) \geq \theta$, that is $2W C(t^*) \geq (2u + 1)T$, at the detector whose segment of that interval's increment, laid out in the ladder's order, contains $\theta - C(t^* - 1)$; there the record ends, whole, everywhere, at that interval. The click is the detector's: it names the detector, never a Node inside it, and is reported per detector [9, 9.25 (2)]; record 1899 [6]. T and C are integers once scaled by 3 and by the pairs' common wall.

The three tests hold for the click: generic, the integers u , T , W and the fluxes, no family name; vector, sums and one comparison per detector, the division's comparison (D), no root; local, each detector's increment is its own Ports', and the record's two integers θ and $C(t-1)$ travel with the record as its residue does. What is not local is the deletion: the record ends at once, everywhere, the one non-local act of the law (the model owner's decision, record 1888 [6]).

Theorem 2 (one quantum, one click). *With the face receiver last on every ladder, every record clicks exactly once.*

Proof. C is non-decreasing and $\theta < T$, while everything that leaves the board offers its whole form at the face; so θ is crossed exactly once, and the deletion at that interval leaves nothing to cross it again [9, 9.25 (2)]. $\square$

Theorem 3 (Born's rule for a detector). *Let the residue u be spread evenly over $\{0, \dots, W-1\}$. Then the probability that a record clicks at detector d is $C_d(\infty)/T$, the detector's share of the record's total inward flux, whatever the flux's time profile, to the grain $1/W$; over the ladder with the face the shares sum to one. A detector's share is the sum of its Nodes' shares.*

Proof. θ is spread evenly over a grid of W points in $[0, T)$. The segments of the increments, one per interval and detector in the ladder's order, partition $[0, C(\infty))$ with the lengths $f_d(t)$, so the click falls at (t, d) with probability $f_d(t)/T$ to the grain $1/W$; summing over t gives $C_d(\infty)/T$. The total length of a detector's segments does not depend on the order within an interval [9, 9.25 (3), (4)]. $\square$

The residues of the law's remainder are equidistributed (COMPUTED), not a permutation, so the counts are binomial about the shares [9, 9.19 (4b)]. Two other forms were weighed and refused: a race in which each Node shouts on its own threshold is not Born's rule (two Nodes receiving the shares 0.9 and 0.1 give 17/18 to the first, PROVED), and exactness needs the chance normalised by the record's remaining norm, which is the increment ladder read cell by cell [9, 9.25 (5)].

Definition 2 (The birth). The reading $\mathbf{E}$ maps a record at a set of cells to its offer there; its transpose $\mathbf{E}^\top$ maps a number back to a record, the real part of a character written on cells [9, 9.13]. A birth is $\mathbf{E}^\top$ at a click, and only at a click. An *emitter* is a body of a massive family with a stock of M excitations: at interval 0 its first excited record is on the board, the body's mode at both levels; excited record k clicks at its own rung on its own centre cell, and at that click it ends and $\mathbf{E}^\top$ writes two records, the born record (one quantum) and, while the stock lasts, excited record $k + 1$ (one quantum); so quanta are conserved at every click, and the emission times are the rungs, the spontaneous emission of the algebra [9, 9.17 (2), (4)]. The born record's two levels are integers of the world written by the generator, on one cell the pair [now, -now], on a line of cells a travelling profile, copied once at the click and never driven afterwards [9, 9.17 (5), (6)]. A *crystal* is $\mathbf{E}$, the arriving record's click at its cells, then the split of the arriving clock into the two born clocks, then $\mathbf{E}^\top$, the birth of an entangled pair with its declared branches [9, 9.13]; record 1879 [6].

Where a tensor enters. In one place only: the pair, a record of fixed rank 2 whose label module is $\mathbb{Z}^2 \otimes \mathbb{Z}^2$, four integer weights on the joint labels, each arm's rows over the Nodes as any record's. It is required only where entanglement is measured, since a product state gives the CHSH sum at most 2 (PROVED: CHSH's bound holds for weights that factor). It is born only by the crystal during the run, never written at interval 0, and two separately born records are never joined; so the generator writes only vectors. The pair's click reads its joint weights $R(o_A, o_B) = J(o_A, o_B)^2$, $J(o_A, o_B)$ the joint amplitude of the outcomes o_A and o_B (apart from the invariant J of Eq. (4)), on the joint ladder of the four outcomes, the one residue choosing the outcome for both arms at once; the click of one arm alone reads the weights summed over the other arm's labels, the partial trace, kept nowhere [9, 9.22 (6), 9.25 (6)]; record 1880 [6].

Theorem 4 (no signalling through the click). *For a pair record, the share of records clicking + at Alice's detector is $|\mathbf{w}_+|^2 / (|\mathbf{w}_+|^2 + |\mathbf{w}_-|^2)$, with $\mathbf{w}_o$ her arm's label state projected on her own axis, exactly in integers and independent of Bob's axis, placement and distance; the same holds for Bob. So the counts at one detector do not depend on what is done at the other, and no information passes through the click, at any speed.*

Proof. Her marginal is $\sum_b R(+, b) / \sum R$, and $\sum_b R(o_A, b) = (a_B^2 + b_B^2) |\mathbf{w}_{o_A}|^2$ by Brahmagupta's identity, Bob's axis being (a_B, b_B) ; the scale $a_B^2 + b_B^2$ is common to both outcomes and cancels in the share. Her click's time is her own arm's flux and her outcome's boundary her own projection [9, 9.25 (6)]. $\square$

What is non-local is Bob's outcome given Alice's, decided by the same residue and the joint weights: the correlation, not a signal. A deletion that spread from Alice's click at one Link per interval would be a local rule at each arm, and every such rule gives the CHSH sum at most 2; so the deletion is at once, the algebra's one non-local act [9, 9.25 (6)]; record 1888 [6].

Theorem 5 (the direction of time). *Given the neighbours' reads, the step $(a_{\text{before}}, r) \mapsto (a_{\text{next}}, r')$ of one row is a bijection of $\mathbb{Z} \times \{0, \dots, 3 \text{ den} - 1\}$ onto itself, with the exact inverse $a_{\text{before}} = \lceil m / (3 \text{ den}) \rceil$, $r = 3 \text{ den } a_{\text{before}} - m$, where $m = \text{num } S_6 - 3 \text{ den } a_{\text{next}} - r'$. The coupled step is two such bijections in a fixed order, inverted in the reverse order. The click deletes a summand and has no inverse.*

Proof. $(a_{\text{before}}, r) \mapsto n = \text{num } S_6 - 3 \text{ den } a_{\text{before}} + r$ is a bijection onto $\mathbb{Z}$ by the uniqueness of the Euclidean division, and so is $n \mapsto (\lfloor n/(3 \text{ den}) \rfloor, n \bmod 3 \text{ den})$ [9, 8.8]; [9, 9.20 (A) 4]. $\square$

The Inside is reversible to the bit on every board and is not symmetric in form, the remainder living on the later level: the rule carries an arrow in its bookkeeping and none in its information. The arrow of time is born at the click and nowhere in the rule. The check that shows it is a property test and never a pin: a planted board with no detector run forward and back under the exact inverse returns bit for bit, and the same board with a detector does not, by the deleted record exactly [9, 9.20].

The families. A family is its vacuum pair $[\text{num}, \text{den}]$ and its label module ($\mathbb{Z}^2$ per arm for the hand, $\mathbb{Z}$ for a family with no hand), and nothing else [9, 9.22 (1)]. From the pair alone follow its rest frequency, $\cos \omega_0 = \text{num}/\text{den}$ (light has none), its operator block on an empty board and its band (Section 3), its stability, whether it can hold a body (only a massive family can), and its remainder's range. A born record's clock is a datum of the birth that makes it, not of the family. A world has at most three families, fixed by one pair per family per Node: light, the experiment's matter and the holder of the tools' bodies (ADOPTED, record 1875 [6]).

The experiment as data. An experiment is a list of what is on the board, and nothing else [9, 9.22 (2)]; records 1875, 1882 and 1885 [6]:

- (i) the board: the extents per axis, and per axis periodic or open, an open axis carrying the face receiver at its border, one border for every family;
- (ii) the families, at most three, each its pair and label module;
- (iii) the material map: which Nodes carry which pair per family, which a coupling (g, G) , which an axis (a, b) , every region a cube by its vertex and edge;
- (iv) the occupation: which bound bodies hold quanta and how many, at which amplitude; which packets are on the board; which bodies carry a stock and birth which family with which born profile; which body births a pair, with the crystal's branches;
- (v) the momentum: every entry carries its total momentum, the wave vector $\mathbf{K}$ of its character, $\mathbf{0}$ at rest; there is no acceleration, a moving entry is written moving at interval 0 and its energy of motion is part of the initial state (record 1884 [6]);
- (vi) the names: the detectors' regions and each birth's ladder, the face receiver last.

Not part of it, and refused at load: a residue, a wheel, a seed written by hand, a rate, a train, a heading, a take, a grace, a table, a per-family border, a ramp.

The initial state. The initial element is

$$\mathbf{v}_0 = \sum_{\text{occupied bound modes}} (\text{the quanta}) \times (\text{the mode's integer profile at the declared amplitude, at both levels}) + \sum_{\text{packets}} \quad (7)$$

with every remainder 0 and nothing else on the board: no light record and no remainder at interval 0 [9, 9.22 (3)]. The modes are those of the composed operator of the body's family, every region in place. The generator is the board's own operator iterated: the power iteration $\mathbf{v} \mapsto (\mathbf{M} + 2\mathbf{1})\mathbf{v}$, $\mathbf{1}$ the identity, applies the law's own read and nothing else and converges to the bound mode from any start, $\mathbf{M} + 2\mathbf{1}$ having nonnegative entries (PROVED by Perron–Frobenius); in integers it is reproducible bit for bit [9, 9.22 (7)]. So the law *steps* the operator, the generator *relaxes* with it, and the click compares and deletes; nothing on the board relaxes.

The load check. The initial element is written once into the experiment's input file, with each occupied mode's clock as a rational a/b , and the engine checks it at load in integers only and says

LAWFUL or REFUSED with the reason: the law and generator the file was made under (a change of the laws forces regeneration); the eigen-equation's residual at every Node of the profile $\mathbf{p}$,

$$|b \text{ num}_i (S_6 \mathbf{p})_i - 3 \text{ den}_i a p_i| \leq b (3 \text{ num}_i + 6 \text{ den}_i), \quad (8)$$

PROVED as the bound for the rounded profile of an exact mode; a/b above the band's top and below 2; every remainder 0; the stability of the composed operator; the rich birth cells; the detectors' cubes; and the refused keys [9, 9.22 (3), (7)]; record 1886 [6]. Recomputation at load is a diagnostic only, since a host's float mode computation is not reproducible across hosts.

3 The massive record kind: the pair per kind, the band, the cone and the block

The law of Section 2 was written for records that hop whole; the law as it stands today gives every record kind one declared pair of integers and lets the same six operations carry mass, a clock and a body [9, chapter 8]. This section states that law as the algebra document states it, before any run: each sentence carries the document's mark, proved there for every grain and every board, carried from the design's proof [11], or computed and not proved; every number is a computation declared before its run [12, 14], and the runs that read them are Section 8's.

The rule with a pair per kind. At every Node a record's row $(a_{\text{now}}, a_{\text{before}}, r)$ is mapped once per interval by

$$3 \text{ den } a_{\text{next}} + r' = \text{num } S_6 - 3 \text{ den } a_{\text{before}} + r, \quad 0 \leq r' < 3 \text{ den}, \quad (9)$$

S_6 the sum of the six neighbours' a_{now} (the Node itself on a folded axis, 0 beyond an open face), then the shift $a_{\text{before}} \leftarrow a_{\text{now}}$, $a_{\text{now}} \leftarrow a_{\text{next}}$, $r \leftarrow r'$. Light is the value $[\text{num}, \text{den}] = [1, 1]$; a massive kind is a value $\text{den} > \text{num}$, and nothing else names the mass. The verbs are (B), one entry num of the declared matrix on the six reads and one entry -3 den on the own row's past; (T), the accumulator translated by that rate; and (D), the division by the wall 3 den with the remainder kept on the row, in the floor form, the remainder in $[0, 3 \text{ den})$, the form whose uniqueness makes the step a bijection (the direction of time, below; [9, 8.8]). The three tests hold: one primitive with one declared pair and no family name, light the value $\text{num} = \text{den}$; the three verbs on the row with no root and no float; the record's own row and its six reads, nothing kept at a Node beyond the row and its remainder.

The band and the pitch. Read on a character $\cos(\omega t - \mathbf{k} \cdot \mathbf{x})$ of a periodic board, Eq. (9) without its remainder holds exactly when $2 \cos \omega = (2 \text{ num}/(3 \text{ den}))(\cos k_x + \cos k_y + \cos k_z)$; with light's band $\cos \omega_l(\mathbf{k}) = (\cos k_x + \cos k_y + \cos k_z)/3$ at the same $\mathbf{k}$ the same line reads

$$\boxed{\cos \omega = \cos \omega_0 \cos \omega_l(\mathbf{k}), \quad \cos \omega_0 = \text{num}/\text{den}}, \quad (10)$$

the massive band as light's band times the pitch's cosine, proved in the algebra document for every grain and every periodic board. At $\mathbf{k} = 0$ the gap is the pitch ω_0 , the rest period $N_0 = 2\pi/\omega_0$ intervals, the rest energy $\hbar\omega_0$, de Broglie's internal clock as the thing compared with; light has $\omega_0 = 0$, a zero mode that is a level and not a clock, so the whole difference between light and a massive record in the algebra is a gap. With $2 - 2 \cos \omega = 4 \sin^2(\omega/2)$ the band is light's form with the pace squared scaled by num/den , one substitution.

The two paces and the exact cone. Near $\mathbf{k} = 0$ light's band gives $\omega_l^2 = k^2/3 + O(k^4)$, the pace $c = 1/\sqrt{3}$ of Section 5.1 reached once more. To second order in ω and $\mathbf{k}$ together the massive band is Klein–Gordon's with the cone $c_m^2 = \cos \omega_0 c^2$, a massive record's limiting pace below light's by $\omega_0^2/4$ (0.56 percent at the declared pair [800, 809], $N_0 = 2\pi/\omega_0 = 42.08$, COMPUTATION); with no truncation in ω_0 , at the band's bottom,

$$\boxed{\omega^2 = \omega_0^2 + c_{\text{eff}}^2 k^2 + O(k^4), \quad c_{\text{eff}}^2 = \cos \omega_0 \frac{\omega_0}{\sin \omega_0} c^2, \quad \frac{c_{\text{eff}}^2}{c_m^2} = \frac{\omega_0}{\sin \omega_0} = 1 + \frac{\omega_0^2}{6} + O(\omega_0^4)}, \quad (11)$$

proved in the algebra document as the expansion at the band's bottom (1.0037 at the declared pair [800, 809], $\omega_0 = \arccos(800/809) = 0.1493$, the family's nominal pitch $\mu = 0.15$, COMPUTATION). So a bound clock in motion dilates by $\gamma_m = 1/\sqrt{1 - v^2/c_{\text{eff}}^2}$ at the massive kind's own cone and not by $\gamma = 1/\sqrt{1 - v^2/c^2}$ at light's: two gammas, apart by about $\beta^2 \gamma^2 \omega_0^2/6$ to second order at c_{eff} (0.19 percent at $k = 3$ on the declared pair, the exact ratio $\gamma_m/\gamma - 1 = 0.188$ percent, the second order 0.186; /4 at c_m , the control), the lattice's own signature (Section 8).

Light against itself. Light's own band, the pair [1, 1], disperses by wavelength and by direction with nothing declared: along a unit direction $\mathbf{n}$ with $|\mathbf{k}| = k$,

$$\boxed{\frac{v_g}{c} = 1 - \frac{(3 \sum_i n_i^4 - 1) k^2}{24} + O(k^4)}, \quad (12)$$

the coefficient 0 on a body diagonal, 1/2 on a face diagonal, 2 on an axis and 4/5 averaged over the sphere (COMPUTATION, the series checked against the exact group pace [27]): a prediction of the form, the quadratic dispersion of light anisotropic on the sky with a cubic pattern and no free coefficient beyond the one scale and the frame, its falsifier a set of bursts in different directions whose dispersion coefficients do not fit one cubic pattern once nature's sensitivity reaches the effect; nature today has bounds and no measured coefficient (row A2 of Section 8).

The block: the foreign object on the board. A body is no new group object and no new verb: a finite set R of Nodes on which the 48 act (the block's shape; a square on a layer), declared as world data like a wall's placement; the rule of Eq. (9) on R with a lowered pair $[\text{num}', \text{den}']$, $\text{num}'/\text{den}' > \text{num}/\text{den}$, a well of the pair in the medium whose own pair is the kind's declaration, the medium's pitch μ ; its clock the bound mode of the map in that well, ω_b , a character of the time translation at $\mathbf{k} = 0$, one element of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ carried by all the cells in step; its motion the characters of the band's surface; its momentum the existing $\mathbf{p}$ in $\mathbb{Z}^3$ and its step the existing verb (T), one integer per axis for the whole block with its remainder, the step k a declaration of kind 3 and the pace of what moves the band's, below the kind's own cone; its push the stress of light's field at its outer Ports; its click the evaluation across R . Nothing is declared in motion: the mode's extent, frequency and regime follow from the pair and the side. The block is no element of the phase circle: its place is in the symmetry group of the cube, the shape of its cells on the board, and its action is on the phase circle, through its table, a shift t of the record's phase applied in the linear form to the record's pair of levels by verb (B), one division per output, no reading of a phase and no register at a Node [14, section 14]; what runs on the circle is the record's phase, as for every record. Two regimes by the side s against the one-Node extent: the well, s small, the mode extending beyond the cells, its clock in motion Lorentz's to first order; the cavity, s comparable or larger, the mode inside the cells, its clock in motion the medium's. These are the foreign objects placed on the GameBoard, the blocks, that stand for the external bodies of an experiment, and every experiment of Section 8 is a world file of such blocks, lamps and detectors.

The motion, one formula and its two limits. The moving block is a resting block of width $\gamma_m s$ along its motion with the same well, read at $1/\gamma_m$:

$$\frac{f}{f_0} = \frac{\omega_b(\gamma_m s, g_w)}{\gamma_m \omega_b(s, g_w)}, \quad \gamma_m = \frac{1}{\sqrt{1 - v^2/c_{\text{eff}}^2}}, \quad (13)$$

f_0 the rest clock's frequency and f the moving one's, both a count between clicks on the block's own record, $\omega_b(s, g_w)$ the lowest mode of the well of side s and depth g_w on the world's own board, a computation per world before its run. The design derives it as the boost of the continuum operator of the band's bottom: the block's declared cells widen in its frame and never contract, and Lorentz is reached as a relation on the characters, no boost being among the 48. On the GameBoard the formula is computed and not proved (to 0.01 percent for $s \geq 12$ on a chain, 0.15 percent on a layer). Its two limits are identities of the formula: in the well limit $f/f_0 \rightarrow 1/\gamma_m$, Lorentz's clock as the thing compared with; in the cavity limit $f/f_0 \rightarrow 1/\gamma_m^2$, the medium's clock; between them the first-order deviation is $\epsilon(\gamma_m^2 - 1)/2$ below $1/\gamma_m$, ϵ the mode's binding depth, the block's carried cells' mark on a bound clock and a prediction. Row 4a's number is this formula at the exact cone on its declared layer world, 0.8116, with 0.8108 at c_m and 0.8132 at c beside it as controls (COMPUTATION [12]).

The coupling of the two kinds. At a cell of R the two records' rows are coupled by one entry of verb (B)'s declared matrix over the other record's two columns, the first difference both ways: the massive row gains $g(a_{l,\text{now}} - a_{l,\text{before}})$, the receive, and light's row gains $-G(a_{m,\text{next}} - a_{m,\text{now}})$, the source, both local to that cell of the block. The scheme is the Euler-Lagrange scheme of a two-point Lagrangian, so it is symplectic and has an exact quadratic invariant J , proved in the algebra document for every board (exact to 10^{-12} on a chain, COMPUTATION); the amplitude form of the coupling is withdrawn as tachyonic, its mass matrix at $\mathbf{k} = 0$ having a negative eigenvalue. The block emits at its mode through the same entry, a lamp at its own rest frequency, and a medium block at rest has the index $n^2 = 1 + Gg/(\omega_0^2 - \omega^2)$ in closed form, a control of the coupling with the cavity's own mode sum beside it and the 4 percent between them open in size.

The detector and the click. A detector is not an element of either group: it is a declared set of Nodes with a rung W , and it counts when the record's motion in its cells, squared and summed, crosses W , the comparison $W \times (\text{the pointer}) \geq (\text{the record's norm})$, integers both, the evaluation (E) of the record on the detector's cells in the detector's own clock. A body is a detector of itself, across its own cells; a light detector is a detector set, a Node or a line of Nodes read as one Port, with a rung on the light record. Light is read by moving the record at the detector's cells, and the detector clicks; the verbs are lossless and the click needs no sink. One click per record, and the record's completion, one gather over all its detectors, is the verbs' only non-local step; nothing else leaves the Inside. Where a world must absorb it declares a take, a damping pair on light's row at the cells of an absorbing object, a declaration of the world and not the law's. What is a measurement stays the reading rule of Section 4.

The direction of time. Given the neighbours' reads, the step $(a_{\text{before}}, r) \mapsto (a_{\text{next}}, r')$ of one row is a bijection of $\mathbb{Z} \times \{0, \dots, d-1\}$ onto itself, by the uniqueness of the Euclidean division, with an exact inverse, the ceiling in place of the floor, a declared variant of the one division: the Inside is reversible to the bit for every grain and every board, and not symmetric in form, the remainder living on the later level. The click is the one deletion of the law, and the evaluation ev with the weight and the rung is many-to-one at every step: the arrow of time is born at the click, and at a declared take, and nowhere in the rule. The world that shows it is a check and never a pin: a train

with no detector run forward and back under the exact inverse returns to its first rows bit for bit, and the same world with a detector does not, by the deleted rows exactly.

Known formulas and predictions. Every row of Section 8 carries (K), a known-formula row, the algebra's identity a textbook formula as the thing compared with and the engine reaching it the control of the engine against the algebra, or (P), a prediction row, a number the algebra gives and no known formula gives; a prediction is not a pass. The eight predictions of the massive kind, each declared before any run with its falsifier in the algebra document [9, 8.9]: the two-pace world, a massive record's limiting pace c_m below c by $\omega_0^2/4$ and its bound clock at c_{eff} ; the bound clock's second term $\epsilon(\gamma_m^2 - 1)/2$; the index in motion; the hop's parametric pump; the two arms held by light alone; the atom's lines at the coupled modes' own frequencies; the click's arrow of time; and the light clock of two bodies, $N_0 = 2L/c$ plus the two ring-ups.

4 From click to click

Everything compared with nature is a count between clicks at a detector or a ratio of such counts; nothing inside the board is compared. This section reads forward: its clicks are those of Definition 1 and its postulates are stated in Section 6.1. This section is the chain from the ring's element read at a click to those counts and to the forms they take, the click frame under the hypotheses (A1) to (A3), each named where it enters; it uses the objects of Sections 5 and 6 by the names defined there. It opens with the six steps between two clicks, the two theorems of the Outside, and the one fact behind the uncertainty relation and Bell's excess; then the chain.

The steps between two clicks. Every comparison in this paper walks one chain of six steps, each with its kind (DETECTOR, GAMEBOARD, COMPUTATION, CONVERSION, HOST; Section 8).

- (1) The birth of a record's rows at a declared source: a declaration, the family's row of integers and the world file's source with its rate against its wall; the wheel's reading u is fixed here (Definition ??, B).
- (2) The flight: the six operations on $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ and the lattice, the phase turning per Link, the split and the rotation by the declared tables, the merge in the ring; the law (P3 the six operations, P4 locality, P5 one Link per interval, P9 the blind flight).
- (3) The arrival at a detector and the click, the one measured event and an action of the law on the state, not a passive read (P6 one read-out, P10 the click axiom, P11 clicks only; the law's rule [33, section 10]): the record's wave runs on the free Nodes and passes through the detector's cells, the detector's pointer accumulates its offer, and the click is the first rung, W times the pointer at or above the record's norm, in the detector's own count (Definition 1); one click per record, its completion one gather over all its detectors, the law's one non-local step; the click is stamped with the detector's own count, the flight of every other record untouched, no back-action; the weight is one quadratic form of the record's element, the evaluation at the roots of unity on the rows and the same form on the pair of levels, then one comparison (DETECTOR).
- (4) The detector's own count between two clicks (DETECTOR), read under the world key `clock_stamp`, declared true in every world of the detector law's launch list (`RUN_LIST.md` and the declarations of `DECLARATIONS.md`, the GO of 2026-09-24). The key is outside this paper's law identity (Section 6.1): this paper's own runs read the interval between clicks as the host's tick at the detector's set (GAMEBOARD [6, record 768]), named as such where it is read.
- (5) A ratio of two such counts, or a count itself (COMPUTATION on clicks; nothing of the board enters).

(6) The identity the algebra states between the ratios, the form of nature compared with. Step (6) is the only place where a formula meets a measurement: the click theorem gives the ratios' identity from the ring, Lorentz's factors under (A1) to (A3), Einstein's step and, in the limit under a shell average, Newton's form; a row of the full record [51, 52] compares the ratio of step (5) with that identity, and with nothing else. Anything read from the board itself, a tick, a presence mean, a body's own record, a shell mean, is a GameBoard diagnostic and enters no comparison; the failures of the full record [51, 52] are named by the step their earlier reading broke (Section 8).

Locality and discreteness Outside, two theorems beside P11. Two properties of the game above the board follow and are not assumed.

Theorem 6 (Locality Outside). *Under (A1), the locality of the verbs on the board, taken here as the hypothesis of the conditional extension, and with the definition of Outside (P6 and P11: nothing leaves the board but clicks), every Outside event is a click and every click moves information one Node at most, so every Outside passage is a chain of clicks between neighbouring places, and the Outside inherits the Inside's locality through the conversion: no reading needs a jump, no velocity read exceeds one Node per interval of the tick, which in a detector's own count is $1/r_D$ Nodes per count (r_D the detector's count per interval of the tick, 1 at rest), and no detector reads at a place no chain of clicks reaches. The exceptions are the model's non-local steps, named: the completion of a record at its click, one deletion everywhere (P6, P11), and the pair's click, one gather of one record from both settings; local Inside, one non-local deletion at the click. Transport on the GameBoard is local, one Node per interval on every Link, and the information a click yields Outside is the information at its one cell, its interval and its count; which cell clicks is decided at the completion, a gather over the record's offers, the one non-local step (P6, Definition 1). The theorem bounds the information a click carries, not the choice of its cell.*

Theorem 7 (Discreteness Outside). *By the click's definition, and under (A1) with the conversion, the counts of one detector form a free abelian monoid and their ratios lie in $\mathbb{Q}$: every Outside quantity is an integer count or a ratio of counts. The least separation of two places read is one Link; the least time a detector times by itself is a pulse to its neighbour and its return, two of its counts at $r_D = 1$; the least step of a moving record is one Node per k counts, so an elementary one-Link segment completed in k counts has the velocity $1/k$, and a velocity read across Δx Links and Δn counts is the rational $\Delta x/\Delta n$, of resolution $1/n$ at a fixed count n and dense as the counts grow; and c Outside is a BOUND, at most one Link per interval on every digital line and $1/\sqrt{3}$ on the fan; Outside reads it as Nodes apart over counts apart with $r_D = 1$. Nothing continuous enters above the board.*

Both are the click frame's, its section 0 and its closing theorem [50], and Theorem 3 of Einstein Outside [37]; neither is an added assumption. What is more than (A1) is not a theorem: that the apparatus is itself inside the law, that emitters and detectors are records of the board and not a second kind of thing, is an assumption about the world or a programme for the law; this paper marks it so and does not use it.

The one fact and its two faces. One fact: a record is read once; the click reads one record into one cell of the ladder and ends it (Definition 1; P6, P11). The uncertainty relation and Bell's excess are that fact seen from two sides; every claim here is the derivation's [35, section 22] or the click frame's [50, section 10], none new. The first face: a record is an element f of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its position the Node its rows end at, its momentum the turn per Link (de Broglie's $\lambda = h/p$ as the declared turn, Section 5.4); the click evaluates f at the N_ϕ -th roots of unity, the discrete Fourier transform on $\mathbb{Z}_{N_\phi}$, so position and phase are the two sides of one transform. On the identification of the circle

as position and its transform as momentum, a hypothesis of the programme and not a derivation (Section 5.4), what holds exactly and with no limit is the support bound $|\text{supp } f| |\text{supp } \hat{f}| \geq N_\phi$ [64] (among its equality cases at $N_\phi = 64$, by enumeration: one phase and all 64 roots, the antiphase pair and 32, the comb of every eighth phase and 8; COMPUTATION) and Weyl's relation $VU = \omega UV$ on $\mathbb{Z}_{N_\phi}$ [66, 67], which is where the law goes: $[x, p] = i\hbar$ as $N_\phi \rightarrow \infty$ with $\hbar = h_q N_\phi / 2\pi = h / 2\pi$ (h_q the energy of one phase step, $h = h_q N_\phi$ Planck's constant in the law's units, an input), Kennard's $\Delta x \Delta p \geq \hbar/2$ in that limit and nothing of the law at finite grain [65, 68], and at finite N_ϕ the entropic bound [69], the click's identity that the bits read and the bits not read of u sum to $\log_2 N_\phi$ per record. What makes it an uncertainty and not a spread is the one read-out: read at a Node the record gives its position and its phase vector is erased, read on the fan's angle it gives its momentum's direction and not the Node; the two are not jointly readable because there is one click per record, and the click acts (step (3) above); the reading is what is uncertain, not the board. The second face: a lamp releases one record with two arms, the joint labels 00 and 11 of weight 1, $\psi = (00) + (11)$ in $\mathbb{Z}^2 \otimes \mathbb{Z}^2$ tensored with its phase in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its coefficient matrix of rank 2, no $\mathbf{u} \otimes \mathbf{v}$ equal to it: entangled by construction; in flight each arm is local and nothing of one arm is written on the other; a setting is the integer rotation U_s with $U_s^\top U_s = n_s I$ (Definition ??, R); the pair's click is one gather of the one record from both settings at its completion, the joint pointer $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$, the ladder cell's weight J^2 , the birth wheel selecting one of four cells through the ladder (Section 5.5). The rows are local and the read-out is not: the non-locality sits in one object, a record of rank 2 read by one gather at two clicks, with no signal between the arms; $E(a, b) = \cos(2\pi(a - b)/N_\phi)$ within the tables' rounding, the marginals exactly $1/2$ (Theorem 11; no-signalling an identity of the integers), $S = 2\sqrt{2}$ before the rounding and $176/64 = 2.75$ at $N_\phi = 64$ (the full record [51, 52], row 1a, against 2.42 ± 0.20 [82]), $181/64$ on the plateau (Theorem 12); Tsirelson's bound [79] a theorem of the operator algebra on the labels; not a local hidden-variable value, because the pair's weight is one quadratic form with cross terms where a mixture is linear (Fine's theorem [78]). What links the two faces is the same one read-out per record: it makes the support bound an uncertainty and denies Bell a hidden joint distribution. Not proven, and said so [35, section 22.3]: that Bell's excess is the same Fourier fact as the support bound. The standard names are the things compared with; the model's own words, the click, the record, the gather and the wheel, name the mechanism.

Inside and Outside, and how a click arrives at Lorentz. This paragraph and the two after it are the conditional extension of the law, the click frame under (A1) and (A2): where (A2) enters, the result is not a consequence of the law as built, which meets (A1) only, and each line names it; the r -free lines follow from (A1), (A3) and the conversion alone. A packet moved Inside, from place to place, and what it says Outside: Inside, the law propagates records as a beam by the algebraic formulas of Section 6.1, where no one measures; Outside, the game above the board, there are detectors and their clicks only, the beam passing to them with amplitudes at the click (P11). So a thing propagates Inside as a beam, rows on digital lines at one Link per interval, and shows Outside as a wave: what a detector reads is the click's weight, the quadratic form of the record's element (Definition 1, the same integer as the square of a phased sum), which adds and cancels as a phased sum at the roots of unity does, so Young's bands and the Mach-Zehnder's ports are the beam's rows read at the click and nothing else (Section 5.4); the wave is the read-out of the beam, not a second thing on the board. A detector's clock is its own record's count of what it emits and receives back, no Node holding a time and an open face without a body having none; a velocity is Nodes apart over counts apart between neighbouring detectors; the read-out is the click (Definition 1). The frame [50]: "To convert the lattice and the momentum inside it into motion Outside, the game above the board, one uses clicks that pass information, a click being the passage of information from Node to

Node at most one Node per interval (A1) and its content an amplitude with a phase that splits each interval between staying and hopping, the mass the staying share (A2); Lorentz of the world is assumed, and what is shown is that this conversion brings it. The symmetry of the conversion, the transformations between click families that preserve (A1) and (A2), is the Lorentz group up to scale, acting on the amplitudes of the passing packet as the Dirac walk's covariance, so that the passage of clicks is Lorentz up to corrections of order m^2v^2 , exact in the continuum limit, a detector's own rate $\sqrt{1-v^2}(1-\kappa^2/6)$ with the mass angle m the spacing over the reduced Compton wavelength; in three dimensions the lattice's anisotropy begins at the fourth order." (A1) and the one line, the relativity of the two directions ($k_{AB} = k_{BA}$), are special relativity's two postulates in the click language, so what is arrived at is Einstein's road walked on clicks, and what the board adds is (A2), the amplitudes that carry $\sqrt{1-v^2}$ into a detector's own rate. The beam law's rows meet (A1), one Node per interval with the flight blind to the phase (P9), and not (A2), their hop a whole-record schedule with no staying amplitude, so their counter ran at the rest rate at every speed, $r = 1$, the rows' reading now history; under the massive record kind the clock's $1/\gamma_m$ is derived from the band's one formula (Section 3; rows 4a, 4b) and the Lorentz group is not, the two one-way Doppler factors read through records, apart by $1-v^2$ under the rows and by the band under the kind, the measurable that decides. No wall of the law and no declared identity enter the reading; Eq. (14) is an Inside formula on the record, read by no detector, whose Outside formulas the conversion gives. What the system gives is a computation Inside whose passage Outside is the clicks, at the quantum level as well; how far Outside represents reality is what the conversion table and the full record [52] show, result by result.

The step beneath and the step above. The Inside step of this chain is not Eq. (??) but the click frame's walk, one amplitude per record that splits each interval between staying and hopping (A2), whose exact invariant is $\boxed{\cos \omega = \cos m \cos \kappa}$, ω the walk's frequency per interval and κ its momentum angle per Link; Eq. (??), which hops whole, carries W only as the declared identity of Eq. (14) (the roads table [52]). The paper's centre is the step beneath, on the GameBoard, Eq. (14) in its frame, the six operations on the record with W its invariant; the direction is one way: the step beneath is the axiom, the conversion of (A1) and (A2) is the map, and the step above is its image, a theorem named as what Einstein wrote. From W , by the conversion, the step above follows: Wc^4 is $E^2 = E_0^2 + p^2c^2$, the gate E_0/E' is $1/\gamma$, a detector's own rate, the drive's fraction is the click's velocity, and the two Doppler factors are the step above read at two detectors. The click frame's section 7 [50] shows the chain with no square declared: the walk's exact invariant $\cos \omega = \cos m \cos \kappa$ gives $\omega^2 = m^2 + \kappa^2 - m^2\kappa^2/3 + O(6)$; the law's own Planck map ($E = h_q n/d = h_A f$, $h_A = h_q N_\phi$ the energy per cycle, $p = \hbar \kappa$ per Link, $E_0 = \hbar m$, the load identity $3hn = QSD$) gives $\boxed{E^2 = E_0^2 + c^2 p^2}$; the whole unit $E' = E/c^2$ gives $W = E_0'^2 + 3\mathbf{p} \cdot \mathbf{p}$ with $3 = 1/c^2 = d$ exactly on the body diagonal (2.954, 2.971, 3.000 by direction, the root's rounding), the corrections of relative order $m^2\beta^2$, exact in the continuum limit. So W is shown to that order as the conversion's identity under (A1) and (A2), no square declared anywhere in the chain [39], and Einstein's relation follows from the Inside step under (A1) and (A2), to second order, an extension the law as built does not meet, its rows hopping whole without (A2); for the rows' dynamics, which hop whole and lack (A2), W remains the declared identity of [6, record 270], the reading's identity and not yet the rows'. In one chain, with nothing of Einstein's put in: the Inside step, the click frame's walk on the record with W its invariant (one amplitude per record that splits each interval between staying and hopping, (A2), whose exact invariant is $\cos \omega = \cos m \cos \kappa$; Eq. (??), which hops whole, carries W only as the declared identity of Eq. (14)); the conversion, an emitter and a click at one Node per interval (A1) and the amplitude's split (A2); the symmetry of that conversion, the Lorentz group up to scale (the click theorem); and its image, the Outside step, a detector's place apart and count apart, whose

most general form above the board is Einstein's step as physics writes it, $d\mathbf{p}/dt = \mathbf{F}$ with $\mathbf{p} = \gamma m\mathbf{v}$, $E^2 = E_0^2 + c^2 p^2$, $d\tau/dt = \sqrt{1 - v^2}$, and the geodesic's first terms ([37], section 3). Lorentz of the world is assumed, through (A1) and the relativity of the two directions ($k_{AB} = k_{BA}$), and what the chain shows is that the conversion brings it: the Lorentz group up to scale is the symmetry of the click families that preserve (A1) and (A2), and Einstein's equations are the form of the Outside step, one step above the board; nothing here derives the Lorentz group from the six verbs, and "nothing of Einstein's put in" says that no factor of Einstein's enters the chain's steps, not that the symmetry is derived: the step's r -free lines (the round trip, the composition, the aberration) are Einstein's exactly from (A1), (A3) and the conversion alone, and its scaled lines are Einstein's if and only if $r^2 = 1 - v^2$, which (A2) gives to order $m^2 v^2$ and the law as built, at $r = 1$, does not. The same road serves Newton (below) and, in the same font, what it does not reach: length contraction, the perihelion's second order, the bending of light, waves, horizons and the field equations [37].

The smallest thing above. What the Outside step is made of follows from (A1) and the click's definition with no run: a click is a Node, a detector's own count and what arrived, and the counts of a family add, so their ratios lie in $\mathbb{Q}$ [50]. Hence the least distance a click can report is one Link, Nodes apart being an integer; the least time a detector times is one count, a pulse and its return two; the least step of a velocity at k counts is one Node per k counts; c is one Link per the least count; and the amount w of a row is the unit that arrives. Every reading Outside is a whole number of these, which is why the world above the board is quantized, a theorem of (A1), (A3) and the conversion (Theorem 3 of [37]; the velocity's quantum not read as such): physics above cannot take this from its own formulas, and the Inside gives it. The speed of light Outside is then the arrival count over the Euclidean distance, $c_D = N_l |D|/T_D$ by a row's age or by a pulse and its return, its anisotropy $1/c_D^2 = 2.954, 2.971, 3.000$ on the heading, the face diagonal and the body diagonal and $1/\sqrt{3}$ in the limit [36].

Newton from the small step. The same way, with five assumptions of the law as built and no (A2) (the crowd read as the age moment, the age wall at coefficient 1, the push, the flight blind, the reading a pulse and its return) and the spreading of K beams over a shell of $N(r)$ Nodes, which is arithmetic and not a law, the fall, the equivalence principle, Kepler's ratio and the clock in a crowd come out Outside. The inverse square comes at the shell mean's rung and the equivalence exactly, M_A cancelling record by record. The register's D3, T and X are their confirmation, and the inverse square's decisive reading is not made [50]. Every result Outside is thus the image of one Inside formula under the conversion, and The conversion table [52] lists the map row by row with the reading that sits on the Outside side.

From an Inside formula to an Outside formula, family by family. The same passage, written three times in full with the attributes of the row tuple of Section 6.1 (Node, direction, age, phase, emitter, content per unit, record, label, amount, multiplicity) named at each step, so that a reader sees which attribute each formula reads and where the conversion takes it Outside. *Light through two slits* (the family **light**, $h = 1$, the circle turns): the lamp puts one record on the board with two rows, one per opening, each with amount 1, multiplicity 2, the lamp's phase p_0 at birth and its own direction on the fan; Inside, each row's phase advances by the declared turn per Link along its digital line, so at the pixel x the two arrive with the phases $p_1(x)$ and $p_2(x)$, the difference $\Delta(x)$ the path difference times the turn (the attributes read: direction, age, phase; the content plays no part, $h = 1$); the merge adds them, $f = x^{p_1} + x^{p_2}$ in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, and an equal pair at $\Delta = N_\phi/2$ cancels; the click evaluates f at the root and squares it, $R(x) = 1024(65536 + C[p_1]C[p_2] + S[p_1]S[p_2])$ (the formulas of Section 5.4) (the attributes read: amount, multiplicity, phase); the rung lays the pixels on the ladder and the birth wheel selects one, the record is deleted. Outside a detector holds only

its counts per pixel: over N_ϕ births the count at x is $b_x - b_{x-1}$, proportional to $1 + \cos \Delta(x)$ within $1/N_\phi$, bright where the two ages differ by a whole turn, which is Young's law with the spacing set by the turn per Link; the reading is the bands' centres 23.5 pixels apart for the exact law's 23.3 (L2b, DETECTOR of the engine before 2026-09-23, history). *A massive body's clock and its fall* (the families **m** and **mass**, $h = 0$, no circle): a body carries its Node, its content M , its momentum $\mathbf{p}$, an integer 3-vector, and its counts table, each count an accumulator (s, r, d) ; Inside, its drive moves it at the pace $|p_a|/(N_l N_w M + |p_a|)$ per axis (N_w the world's width) where the axes' fires do not coincide, one Link per interval in all (the attributes read: content, momentum), its clock counts at the rate $1/(1 + a_\tau n/d)$ with a_τ the age moment of the crowd's rows at its Node (the crowd's attributes read: age, amount, direction; the body's: its counts), and the push reads the label flow $\mathbf{a} = \sum \text{amount} \times \mathbf{u}_D$ of the arrivals into $\mathbf{p}$ (the attributes read: amount, direction, content). The conversion is a lamp on the body and a detector at rest: the lamp's births are the body's own counts, the detector counts their arrivals with its own count, and Outside there are only the two counts and the births' Nodes. So $1 + z$ is the ratio of the two counts. The ratio of two lamps' shifts at 3 and 6 Links is the clock's form at two distances, 1.907 for the pin 1.909 (series T, DETECTOR; read before the generic entry of 2026-09-22, not readable under the law as it stands, being re-read in the weak field). The fall is the birth's Node per count read at a line of detectors, the same Node on 138 of 139 births with the held mass four times (D3, DETECTOR), the equivalence read with no mass in the reading. *The pair* (the choosers **sa** and **sb**; the pair's record carries the labels $\{0, 1\}$ with integer weights on two arms): Inside, a setting a acts on an arm's label pair by the integer matrix U_a (the attributes read: label, amount, phase on each arm; the record's identity joins the arms), the joint weight is $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$ and the ladder cell's offer $R = J^2$, Theorem 11; the conversion is the one gather of the record from both settings at the completion interval (P6), the rung over the ladder's four cells and the wheel's choice. Outside the two detectors hold four counts per setting pair: $E_{N_\phi}(a, b) = 4c_{++}/N_\phi - 1$ and $S(N_\phi)$ from them, Theorem 12, with each party's marginal $N_\phi/2$ exactly; the reading is $S = 2.75$ at $N_\phi = 64$ and $32/64$ in every bin (series L, DETECTOR of the engine before 2026-09-23, history). In each of the three, Outside holds counts and Nodes and nothing of the attributes themselves: the phase, the amount and the multiplicity of one record never pass singly, the content passes as a wall's scale, and what passes is what the click compares. The conversion table, in the paper's records file [52], names, row by row, the family and the attributes each Inside formula reads.

What the Inside step gives Outside that the continuum cannot state. Formulas of the six verbs taken Outside by the conversion, each with the click that reads it and its kind; none of them is a formula of the continuum; the full lists, each candidate under the three tests with its reading, are [40] for quanta and bodies and [41] for light. (i) The quantum of the Outside step, stated above (one Link, two counts, $1/k$ for a one-Link segment of k counts and a rational $\Delta x/\Delta n$ across more; Theorem 3 of [37]), read as a least step in series S's face clicks 369 and 345 (under the hypothesis covariant-readings-v1, a hypothesis's reading, history) within two of the flight table's and D3's 138 of 139 births (history) at the same Node (DETECTOR); the velocity's quantum and the pulse's two counts not read as such. (ii) The pair's sum at the CHSH labels is the exact rational the rung gives, $|S(N_\phi) - 2\sqrt{2}| \leq 8/N_\phi + 0.0444$, above the bound at $N_\phi = 16$ and 32 ($S = 3$), the plateau $181/64$ from 512 through 8192 below it by 3.02×10^{-4} and $5793/2048$ above at 16384 (Theorem 12; the frame's section 10); measured at seven grains, 64 and the six from 512 to 16384 (the $S(N_\phi)$ figure of the engine before 2026-09-23, history [52], DETECTOR). (iv) The equivalence's constant: the clock in a crowd and the accelerated detector read one shift, $\delta k = (nS/d) (gY/c^2)$, $nS/d = 16$ a DECLARATION (an input of kind 2), Einstein's gY/c^2 exactly when $nS = d$ (times the grain's $\tau_{LC} = 0.993$ on a heading, 1 in the limit of every direction), a condition on an Inside declaration

and not a result; the register's clock and crowd worlds declare $nS/d = 16$ (a DECLARATION of the world files, an input), so the condition is not met by them, the form's readings are series T's 1.907 and series X's k (DETECTOR; T read before the generic entry of 2026-09-22, not readable under the law as it stands, being re-read in the weak field), and the constant's reading is NOT MADE ([37], II.10). (v) Light's bending: on the law at head, half of nature's, by the time part alone, -1.993 pixel at $\gamma_{\text{PPN}} = 0$ on a coupled world (DETECTOR of the engine of 2026-09-22, history; NOT PREDICTED under the law as it stands; the pin met then [30]), a factor 2 against nature's $1 + \gamma = 2$, 1.75 arcseconds. Series K's 0.000 pixel (DETECTOR) the reading of the worlds that declared the coupling away, history, before the generic entry of 2026-09-22. Under the key **optical**, the flight in the age wall's set with a declared c_f , the delay $c_f(nS/d)(GM/c^3) \ln(4r_1r_2/b^2)$ and the bending $\alpha = 2c_f(nS/d) GM/(c^2b)$, Einstein's at $nS = d$ and $c_f = 2$, NOT COMPARED, the route chosen ([37], II.11). The chain, algebra first and the run after (Section 5.3; the full record [51, 52], row 13): the step algebra's ring on the law as built, $C_{\text{ring}} = 5.12$ at $b = 6$, the M/b form recovered and 4 not reached (COMPUTATION [44]); under the one constant 3.65 to 3.92 against 4, $2c_f$ within two stated factors (COMPUTATION [45]); the run of 2026-09-22 under **flow_link**, the $\gamma = 0$ ring at $b = 6$ inside its four pins (DETECTOR) and, on the split ladder, every ring read inside its pins (the deciding pin 0.731 against 0.731 ± 0.025 , DETECTOR), the constant through the stated factors outside the one-grain pin by 2.40, 1.07, 1.40 grains at $\gamma = 1$: DOES NOT CLOSE within the grain [47]. The conjecture, labelled so: on the one constant the deflection coefficient is $2(1 + \gamma)$, γ the declared input, conjectured from the algebra and not read (Section 5.3). The families of the register, each with its declared integers, its algebraic object and the click that reads it, are the families table [52].

The limits, and what does not return. The content and $\sum w^2/\mathfrak{m}$ are conserved exactly, the offered norm only within the tables' rounding. The covariant readings [6, record 270] read one Inside formula Outside on one axis, series S against its pins: a body's record carries the exact square of its energy,

$$\boxed{W = E_0'^2 + 3\mathbf{p} \cdot \mathbf{p}, \quad E_0' = E_0/c^2 = N_l N_w M, \quad c^2 = 1/3} \quad (14)$$

compared and never rooted, with E' the largest integer whose square is at most W and every count of the body gated by E_0/E' ; multiplied by c^4 it is the energy-momentum relation of special relativity (series S, the conversion table [52], inside their pins). Eq. (14) is a declared identity of the record's representation in the law as built, not derived from the rows' dynamics; under (A2), an extension the law does not meet, W is shown to second order (the step above, this section). The hypotheses beside the law are on the tree [31].

5 From the algebra, the physics

What follows from the algebra is stated first in one paragraph, each item with its word (exact; recovered in a limit; reached under named hypotheses; conjectured), and then subsection by subsection with the proofs, the runs that read each form after a detector, and the words kept apart.

What is proved. Derived Inside, exact on the GameBoard, no experiment needed: the group of the six Ports and, within it, the group of order 24; the books' conservation and the lattice form of the continuity equation; Gauss's law of a free family's flux under its world condition (the row form's theorem, its record); the split's isometry and the injectivity of the interval on the amplitudes for a fixed event history; the pace's bound, one Link per interval on every line, and $c = 1/\sqrt{3}$ the band's pace at $k \rightarrow 0$ on every direction, its dispersion declared; the click's weight a positive quadratic

form of power 2 under the parallelogram axiom; the exact marginals of a pair; the CHSH sum as an exact function of the grain and the tables; Planck's and de Broglie's relations as identities under the declared dictionary. Recovered in the limit of the grain, with a rate: Born's rule to $1/N_\phi$ per cell and the cumulative rungs to $1/(2N_\phi)$ with the exact tables (to the tables' rounding with the declared ones, Section 5.4), Tsirelson's bound to $8/N_\phi$ plus the tables' 0.0444 ($2\sqrt{2}$ in the joint limit of N_ϕ and N_t , the tables' scale), Young's spacing, the attained value of c , the continuity equation's differential form. Recovered under an average that covers the shell, with the condition in the claim: Newton's inverse square, from the clicks (Section 4); Coulomb's inverse square, the retarded potential, Poisson's equation and the clock's $1/r$ form were derived from the push of the rows that hop and read on the engine before 2026-09-23, history [52]. Each names the known form the algebra recovers (Gauss's, Newton's, Coulomb's, Born's, Tsirelson's, Planck's, Einstein's); the paper's own are the integer road, the finite- N_ϕ values (Born's bound $1/N_\phi$, the plateau $181/64$, the fixed-table limit), the quadratic read-out characterization on $\mathbb{Z}_{N_\phi}$ (under the parallelogram-law axiom), the measurable $1 - v^2$ and the entropy identity of the click (an identity of the wheel); the value $1/\sqrt{3}$, Tsirelson's limit and the dictionary's identities claim no novelty. Conjectured from the algebra and not read, labelled so wherever they appear (Sections 5.3 and 8.2, the forcing ledger [52]): the bending's coefficient $2(1 + \gamma)$ on the one constant, and the atom's ladder $a_j/a_i = (j/i)^2$, $E_j \propto 1/j^2$, with Balmer's $27/20$ in the shell mean's limit; the ring's run reads its rule inside its pins and its constant outside the one-grain pin, and the ladder is computed from the algebra, not read on the board. After a detector every reading sits inside an expectation written before the run; against nature, the paper's list of Table 1 is closed in the algebra before any run, every pinned reading a blank until its run is read and merged, and none is read yet on the engine as it stands; the old count of the engine before 2026-09-23 is history [52]; the hypotheses beside rows 6, 13 and 14 change no verdict of the law (the full record [51, 52]). What no rule of the law forces, with the check that decides each. Newton's inverse square, closed from the clicks under its hypotheses (Section 4); its old reading (D3) and Poisson's law after a detector are history [52]: decided by a run under the law as it stands, pinned before it (Section 8). The Lorentz factor, the law meeting (A1) and not (A2): decided by the ratio of the two one-way Doppler factors through records, $1 - v^2$ under the law and 1 under Lorentz, and by the split of a row between staying and hopping (A2), a hypothesis not opened. The bending of light, the flight blind to the crowd: decided by a rule under which a row's wall reads the crowd's age moment (a hypothesis, its coefficient an input) against nature's 1.75 arcseconds. The masses' values, every rule linear in the content and every equal split a fixed point: decided by a nonlinear closure on the amounts under its own identity, the family table an input until then. The click's harmonic constants and the circle as position (a hypothesis, Section 5.4): decided by the least-rank click, an axiom of the apparatus pinned by the two-slit period and Malus at 22.5 degrees, and by the single opening against its pin (row 10). And a CHSH measurement at about 6×10^{-5} , deciding $181/64$ against $2\sqrt{2}$ under the five-sigma rule.

5.1 Geometry and symmetries: the directions and $c = 1/\sqrt{3}$

The six Ports and their group are Theorem 1 of Section 2; the symmetries of the law and the pace follow from them.

The symmetries. The law is invariant under the translations of the box and of the interval, and under the 48 signed permutations of the axes, the group of the octahedron of the six Ports and of the cube ($3! \times 2^3$; the 24 of determinant $+1$ its rotations, the 24 of -1 its reflections, told apart by the hand, a pseudoscalar of the group), up to the law's two declared ties, the digital line's axis order and the collision's Port order ([2], note 39 and section 4); under the circle $\mathbb{Z}_{N_\phi}$ exactly on the

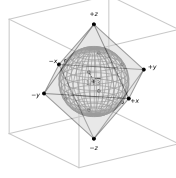


Fig. 2 The Nodes one interval from a Node: the six neighbours are the vertices of the octahedron $|x| + |y| + |z| \leq 1$, the causal front of one interval; its inscribed sphere, of radius $1/\sqrt{3}$, touches the eight faces on the cube diagonals and is the rows' pace c (Proposition 2); the faint cube shares the 48 signed permutations of the axes. Drawn by `octahedron.py` from the definitions; no run.

turns between events and at the click only up to the tables' rounding; and under the permutations of the families, each with its columns of the family table carried along [8]. A boost, which mixes a step in space with a step in time, is not among the 48: Lorentz's symmetry is not a symmetry of the GameBoard but of its clicks Outside, under the amplitude split (A2) that the law as built lacks, to second order in the velocity, the GameBoard's corrections from the fourth (Section 4). The 48 are the GameBoard's stand-in for the rotation group; angular momentum is conserved only up to the GameBoard's grain, and no Noether theorem for the map is claimed [6]. The name of the program counts them.

History. The beam law's flight table, the runtime before 3a9a7109 (two rules, one Link per interval and the digital line of a direction, the octahedron of the six neighbours the causal front, the ground of Proposition 2), is in the records file [52], history: under the law as it stands light is the record kind $[1, 1]$ whose band disperses at k^2 (row A2), and c Outside is a bound (Theorem 7).

Proposition 2 (The pace). *For every primitive direction D , $S_1 N_l \leq T_D$: the Manhattan pace $S_1 N_l / T_D$ is at most one Link per interval on every digital line, and $S_1 = \sqrt{3} |D|$ exactly when $|D_x| = |D_y| = |D_z|$. A pace v that is the same in every direction and at most one Link per interval on every line satisfies $v \leq |D| / S_1$ for every D , hence $v \leq 1/\sqrt{3}$, the radius of the octahedron's inscribed sphere; and the table's Euclidean pace obeys $1/\sqrt{3} \leq N_l |D| / T_D < (1/\sqrt{3})(1 + 1/T_D)$ for every D .*

The proof (Cauchy–Schwarz against $(1, 1, 1)$), the Courant and lattice Boltzmann comparisons and the ground of the bound are in the records file [52], history.

So $c = 1/\sqrt{3}$ Links per interval is the largest isotropic pace the two rules allow, and the table sits at it; that it sits at the supremum and not below is the flight's declared wall $T_D = \lfloor \sqrt{3|D|^2 N_l^2} \rfloor$, the design's third statement, a rule chosen among few (P9; the derivation's 24.1, rows 7 and 9), not a consequence of the two; given the wall the value is the flight operator's norm, so the value rests on locality, straightness and the declared wall. Stated exactly: the two rules give $c \leq 1/\sqrt{3}$ (one Link per interval allows the pace 1 on a heading and only $1/\sqrt{3}$ on a cube diagonal, three Links for the displacement $\sqrt{3}$), and what is derived is the largest uniform pace the GameBoard allows under them; why the rows realise it, and what a detector in motion reads of it, are not derived here (Section 4). The anisotropy is the table's rounding, below $1/T_D$ in every direction and so of order $1/(\sqrt{3}N_l)$ (the full record [51, 52], row 5a, the bound on N_l ; the bound assumes the rounding's anisotropy is of order $1/N_l$ at every N_l , as computed at 64; an upper bound alone gives a sufficient N_l , not a necessary one); nothing measured enters the value. Stated with its ground: the pace of a direction, $N_l |D| / T_D$, isotropic within $1/T_D$, and the Manhattan bound $S_1 N_l \leq T_D$ are exact in

integers; the isotropic bound $1/\sqrt{3}$ is a theorem about a pace the same in every direction, attained on the eight diagonals, above on every other registered direction, and a limit of the grain N_l .

5.2 Conservation, the flux and the forces

The books. Every message released is on exactly one line of the books at every interval, released equals in transit plus absorbed plus escaped plus cancelled, per family, in amount and in content, exact at every interval and at every operation (P3, P5): the carry is a bijection $s \mapsto (e, s - ed)$ that loses nothing, and the split and the rotation are integer matrices whose conjugate transposes invert them (Theorem 8). From the books the continuity equation of the content follows exactly on the GameBoard (the derivation's 25.4 [35]); it is the first row of the ledger. A paid message carries its label out of its emitter at birth and into its reader at its end, so for paid messages momentum is conserved and the third law holds message by message; a free message, a field's row, takes no recoil at birth and pushes its reader by its label times the reader's content. For a free family the third law is a symmetry of two readers, body A pushed by B's field and B by A's, the two pushes $-M_A \mathbf{a}_B$ and $-M_B \mathbf{a}_A$ (M the content, $\mathbf{a}$ the acceleration the other's field gives) equal and opposite, and not a balance per message; the momentum lines of the free books are a report of the GameBoard, and the momentum of the bodies and the field together is not booked as conserved (the derivation's 9.1 and 12.4 [35]).

Gauss's law, exact. The walk is a translation: a free unit released inside a closed surface that no periodic axis crosses, with no absorbing reader inside it and within the unit's lifetime, crosses the surface outward once more than inward (a non-convex surface it may leave, re-enter and leave again, the signed crossings summing to one, the crossing counted with the sign of the Port's outward normal; no splitter, absorber or cancelling pair inside the surface; no collision acts on a fan or on lone beams), so once the front has passed, the net amount crossing the surface per interval equals the release inside it for a constant release, exactly, and for a varying release the crossings at the interval t equal the releases at the retarded intervals, direction by direction. The crossings of an interior surface are the host's count (GAMEBOARD, a diagnostic); a face's border clicks are a detector's (DETECTOR). Proved from the definitions under those conditions; no detector reading of it is registered.

5.3 Light past a held mass: the time part alone

Light: the time part alone, no optical metric. The law at head bends a light row past a held mass by the time part alone, half of nature's, a reading of the engine of 2026-09-22, history, NOT PREDICTED under the law as it stands; its readings, the ring's run and the delay's chain are in the records file [52] (row 13, history).

5.4 Measurement and the quadratic form

The worlds of this section have no body: the law of Section 6.1 on rows alone, and the measurement is the one threshold that is read out. Every interval applies three of the six operations to the state, the translation (the flight, the phase, every count and the birth), the multiplication by a declared integer matrix (the split and the rotation) and the sum in the group ring (the merge, where opposite phases cancel); between clicks the rates are constant, so the map has a closed form and every count of Section 8.1 was computed from it before its run. The click is the one place with a fourth operation: the apparatus sums a record's rows through the tables and squares the sum, the norm, and only then applies its threshold to the birth phase u .

The row and the record, the update rules of one interval and the click are Definitions ??, ?? and 1 of Section 2, one click per record by that Definition.

The two theorems of the interval. Two theorems of the linear block are stated here and proved in the paper's records file [52].

Theorem 8 (The split is an isometry whose conjugate transpose inverts it). *The split $(w, \mathbf{m}, p) \mapsto ((wa_i, \mathbf{m}A, p+t_i))_i$ preserves $\sum w^2/\mathbf{m}$ for every nonzero integer vector (a_i) , and the conjugate transpose of the table (the same weights with the turns reversed) applied to its outputs, followed by the normal form, returns the input up to the scaling $(kw, k^2\mathbf{m})$ with $k = A$. The rotation satisfies $U_s^\top U_s = n_s I$ exactly with $n_s = C'[s]^2 + S'[s]^2$, and multiplies a record's norm by $n_s/65536$: an isometry only up to the tables' rounding (65705/65536 at $N_\phi = 64$, $s = 1$).*

Theorem 9 (The interval is injective between clicks). *Let Q be the quotient of $\bigoplus_r M_r$ that forgets the age (rows differing only in their age identified), and fix the event history of the record, the intervals and Nodes of its births, splits, rotations and re-emissions, which the record keeps. For an interval without a click, an end at a face or the border, or a collision, and for splitters whose joint table B (one column per input, the entries $a_i x^{t_i}$ in the group ring) satisfies $B^H B = AI$, the balanced splitter with the turn $N_\phi/4$ among them, $M \circ R \circ S \circ F$ induces an injective $\mathbb{Z}$ -linear map of Q into itself: no rule of the interval deletes a weight, a multiplicity or a phase, and the state at any interval between two clicks, with the history, determines them at the birth. The age is not a coordinate of Q but a function of the history, the current interval less the interval of the row's last event; a split restarts it at 0 by the definition of age (Definition ??) and forgets nothing, since the interval of the split is in the history; the click is the one deletion. A collision, where it acts, is a bijection of the GameBoard's states that is not linear; none acts in the worlds of this paper. The theorem is of the rows' linear block; a body's drive, whose later coincident fire is lost under the engine's rule (P5, Section 6.1), is outside it and is not injective in that case.*

The theorems under the record kind. Theorems 8 and 9 are theorems of the rows' linear block. Under the record kind of Section 3 the same tables act on the record's pair of levels by the linear form [9, 1.7]: the turn by t is the linear map $(a_{\text{now}}S[k+t] - a_{\text{before}}S[t])/S[k]$, inverted by the turn by $-t$ with the remainder kept, and it carries the character to the character with its amplitude unchanged; so no interval between clicks deletes the amplitude or the phase, and the isometry is of the amplitude, exact in the reals and within the tables' rounding in the integers. That inheritance is proved there by one identity and is not restated here. Theorems 10, 11 and 12 are theorems of the click, of its weights and its counts (Definition 1), which the record kind leaves as they are; they hold under the law as stated.

So a record's norm is conserved exactly by S and by the flight, and by R and by every read through the tables only within the tables' rounding (Appendix A). The normalisation by C_K in (??) makes the click's probabilities sum to one at completion; it does not restore the norm along the way, and the paper claims exact unitarity nowhere.

The click's square as a positive quadratic form: a quadratic read-out characterization. The row of the forcing ledger [52] on the click's weight is written out here from the derivation's section 6.5 [35]; the register's checks are series L's clicks (Section 8.1). The click's square, the norm of a record's summed pointer, is the one operation of the law beyond the translation, the wall, the declared matrix and the addition, and the place where the Born rule enters. Asked: whether the operations of Section 6.1, with one hypothesis on the reading, force the reading at a click to be that square, or what weaker statement holds and what the register pins.

The objects are those of Section 2: the record's element f of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, its pointer $\text{ev}(f)$ and the function R of a cell's element.

The hypotheses. (a) The phase rotation, $R(xf) = R(f)$: a rotation of the record's phase by one step changes no weight (the rungs take no u ; u only selects among them), an operation of Section 6.1. (b) The balanced splitter conserves the reading for every two inputs, $R(x^{N_\phi/4}f + g) + R(f + x^{N_\phi/4}g) = A_2(R(f) + R(g))$ for all f, g (the $(1, 1)$ split with the quarter turn on the reflected row, A_2 its multiplicity factor): a hypothesis on the reading, stronger than Section 6.1's split, which is an integer matrix on rows and says nothing of what a detector reads under it; the register checks it only on the registered pairs (the totals 64/0 and 32/32 of Section 8.1). (c) Counts, $R(f) \geq 0$. (d) The empty reads zero: $R(0) = 0$ on the zero count vector. (e) Some input reads: R is not identically zero. Without (d) the hypotheses admit a detector that always reads 1, which satisfies (a), (b) with $A_2 = 1$ and (c) and is no quadratic form; without (e) they admit the zero detector, under which (b) holds for every A_2 ; so (a) to (c) alone do not force the square. (d) and (e) are what the engine does (the Gram form on the empty record is 0; a click needs a live unit) and what the proof's first step uses ([6, record 340]; the derivation's 6.5 restated with them). Not assumed: continuity, a dimension, the form of R on one row, or Theorem 8's $A = \sum a_i^2$.

Theorem 10 (The quadratic read-out on the phase lattice, a characterization under the parallelogram law). *Under (a) to (e), with N_ϕ a power of two, $A_2 = 2$ and*

$$R(f) = \sum_{j \text{ odd}, 1 \leq j < N_\phi/2} c_j |\sigma_j(f)|^2, \quad c_j \geq 0, \quad \sigma_j(f) = \sum_p f_p e^{2\pi i j p / N_\phi}, \quad (15)$$

the σ_j the Galois conjugates of ev , $\sigma_1 = \text{ev}$. Every such R is a positive quadratic form on the lattice, homogeneous, $R(af) = a^2 R(f)$; it conserves the total at every declared split of one row or of two recombining inputs exactly if and only if the multiplicity factor is $A = \sum a_i^2$, and at every label rotation with the factor $C'^2 + S'^2$, C' and S' the half-angle tables of Definition ???. Two rows ax^p and $bx^{p+\Delta}$ at one Node, Δ their phase difference in steps, read $R = C_\Sigma(a^2 + b^2) + 2ab\mathcal{K}(\Delta)$ with $C_\Sigma = \sum c_j$ and $\mathcal{K}(\Delta) = \sum c_j \cos(2\pi j \Delta / N_\phi)$, $\mathcal{K}(0) = C_\Sigma$, $\mathcal{K}(N_\phi/4) = 0$, $\mathcal{K}(N_\phi/2) = -C_\Sigma$ for every admissible R ; the Born form $|\text{ev}(f)|^2$ is $c_1 = 1$ and the other $c_j = 0$.

Proof. Sketch; the derivation's 6.5 [35] gives the steps in full under (a) to (e). (i) $g = 0$ in (b) with (a) gives $2R(f) = A_2(R(f) + R(0)) = A_2 R(f)$ by (d), so $A_2 = 2$, one f with $R(f) > 0$ sufficing, and (e) supplies it. (ii) Replacing g by $x^{-N_\phi/4}g$ in (b) and using (a) and $x^{-N_\phi/2} = -1$ gives $R(f - g) + R(f + g) = 2R(f) + 2R(g)$, the parallelogram law on the abelian group $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$. (iii) The parallelogram law alone makes R a quadratic form: on a lattice the argument of Jordan and von Neumann [70] needs no continuity, and $R(f) = B(f, f)/2$ with B a $\mathbb{Z}$ -bilinear symmetric form, the restriction of the real form of a symmetric matrix $\mathbf{G}$, the click's Gram matrix (not the fan's grain N_θ of Section 6.1). (iv) By (a), $\mathbf{G}$ commutes with the signed permutation of the basis that multiplication by x is, whose eigen-functionals are the σ_j , j odd; so $\mathbf{G}$ is diagonal in that basis with real entries paired under $j \leftrightarrow -j$. (v) A negative c_j would make the form negative on an open cone containing lattice points, against (c). (vi) The conservation at a general split and at a label rotation follows from homogeneity, (a) and the cancellation of the cross terms between the two outputs; for two inputs recombining at one split it needs $\mathbf{B}^H \mathbf{B} = \mathbf{A} \mathbf{I}$ per character, which the balanced table meets, the declared splitter being that case. (vii) Two rows: $|\sigma_j(a + bx^\Delta)|^2 = a^2 + b^2 + 2ab \cos(2\pi j \Delta / N_\phi)$. $\square$

What is derived and what is not. Derived from the update rules and the axiom (b): the reading is a positive quadratic form of the record's element, forced by the rotation, the balanced

splitter's conservation and the counts alone, with no continuity and no dimension (Gleason's theorem [63] needs dimension three; the lattice version does not, the parallelogram law being granted by the balanced splitter); the multiplicity rule $A = \sum a_i^2$ and the Gram form are derived under the parallelogram-law axiom (quadratic, positive, on the lattice), not quadraticity itself; the physics is the built form, the record's motion squared summed across the cells, which satisfies the axiom; and the cross term $2ab\mathcal{K}(\Delta)$ is forced at the click. Not reached by the algebra: the constants c_j , the detector's response to the harmonics of the phase; a single harmonic j is the Born form under the relabelling $p \rightarrow jp$, a mixture a different law, the flat mixture the phase-blind detector, which interference excludes (the full record [51, 52], row 2a; the built click). As N_ϕ grows the admissible readings are the same family, the cone of the even positive-definite functions on the circle with odd harmonics only; no N_ϕ and no continuity forces $c_j = [j = 1]$, and the fundamental is what the register reads.

Interference, by the formulas alone. Every phenomenon of this paper is read the same way: Outside, what a detector counts; down to Inside, the rows, their phases, the merge and the evaluation; back Outside, the click's counts over the births against the reading. Two slits: one record is born with two rows of amount 1, one per opening, $m = 2$; each row's phase advances one turn per Link; at pixel x the two arrive with phases $p_1(x)$, $p_2(x)$, their difference $\Delta(x)$ the path difference times the turn; the merge adds them and an equal pair with $\Delta = N_\phi/2$ cancels to no row; the click evaluates the sum and squares it, the offer $R(x) = 512 [(C[p_1] + C[p_2])^2 + (S[p_1] + S[p_2])^2] = 1024 (65536 + C[p_1]C[p_2] + S[p_1]S[p_2])$ up to the tables' rounding, proportional to $1 + \cos \Delta(x)$; the rungs b_x of Eq. (??) lay the pixels on the ladder, the click is the pixel under which u falls, and over N_ϕ births the count at x is $b_x - b_{x-1}$, Born's to $1/N_\phi$, with the exact tables. With the declared tables at $1/256$ the weights vary with the birth phase within the tables' rounding (hypothesis (a) of Theorem 10 holds only to that rounding), so the count over the births departs from the rung difference by up to that rounding: two offers of equal ideal weight at the phases u and $u + 1824$, $N_\phi = 8192$, read 4104 and 4088 for 4096 each, a departure of $8/N_\phi$ (a computation from Eqs. (??) and (??), not a run); the built click's bound is the tables' rounding, the theorem's $1/N_\phi$ its ideal. The spacing is the law's own: $\lambda = cN_\phi d/n$ Links, one turn of the phase per age per fringe, Young's $\lambda D/s$ only in the paraxial limit (rung 2, the fan) [36]; the bands' centres are 23.5 pixels apart for the exact two-path law's 23.3 (L2b, DETECTOR of the engine before 2026-09-23, history). What passes is a pixel with Δ near 0, a wide rung; what does not is a pixel with $\Delta = N_\phi/2$, the rows cancelled, a rung of width 0 where no click can land; one slit is one row per pixel, nothing cancels, a flat count within the fan's grain and no dark pixel; the Mach-Zehnder's two ports are the same algebra at two Nodes, the bright port 64 of 64 clicks and the dark port 0 (row 2b; Appendix B).

Planck and de Broglie, as identities. Under the declared dictionary the release's cost rule reads as $E = h_q s = (h_q N_\phi) f$ and the turn rule as $\lambda = h_A/p$: Planck's and de Broglie's relations are identities of the update rules with one constant, $h = h_q N_\phi = h_A$, an input (the forcing ledger [52]); no detector reading of them is registered.

The uncertainty relation. On the identification of the circle as position and its transform as momentum, a hypothesis of the program and not a derivation ([35], section 22), the support bound $|\text{supp } f| |\text{supp } \hat{f}| \geq N_\phi$ [64], the entropic bound [69] and Kennard's form $\Delta x \Delta p \geq \hbar/2$ in the limit [65, 68] are the transform's mathematics on $\mathbb{Z}_{N_\phi}$ and nothing of the law; the entropy identity of the click, bits read plus bits not read of u equal to $\log_2 N_\phi$ per record, is an identity of the wheel. What makes it an uncertainty is the one read-out: a record read at a Node gives its position and loses its phase, read on the fan's angle it gives its momentum's direction and not its Node; the same one

read-out is what denies Bell a hidden joint distribution (“The one fact and its two faces”, Section 4; [50, section 10]).

5.5 Bell and CHSH: from weights to counts

The correlation at the settings (a, b) is $E = (c_{++} + c_{--} - c_{+-} - c_{-+})/N_\phi$ over N_ϕ births, one per u , and the CHSH sum is $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$ at the labels $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$. Two theorems follow from the definitions, exact on the GameBoard.

Theorem 11 (Exact count marginals). *For a pair with labels $\{0, 1\}$ of equal weight and settings a, b , let $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$ and $R(o_A, o_B) = J^2$, the four joint cells in the declared order of Definition 1. Then the first party’s outcome is $+$ exactly for $u < N_\phi/2$ and $-$ for $u \geq N_\phi/2$, for every (a, b) and every N_ϕ ; and the second party’s count of $+$ is exactly $N_\phi/2$ for every (a, b) unless $2N_\phi R(+, +)/C_K$ is an odd integer (a tie of the rung), in which case it is $N_\phi/2 + 1$. No tie occurs for any setting pair at the twelve grains from 8 to 1024 of [5], nor at the CHSH labels for any multiple of 8 up to 4096.*

Proof. Write $n_a = C'[a]^2 + S'[a]^2$. Since the rows of U_b are orthogonal on the integers, $\sum_{o_B} J(o_A, o_B)^2 = n_a n_b$ for each o_A , so $C_K = 2n_a n_b$ and the coarse cumulative after the first party’s $+$ is $C_K/2$: $b = \lfloor (N_\phi C_K + C_K)/(2C_K) \rfloor = \lfloor (N_\phi + 1)/2 \rfloor = N_\phi/2$. For the second party, $R(o_A, o_B) = R(-o_A, -o_B)$ (the rows of U_s are each other’s quarter turn), so with $x = N_\phi R(+, +)/C_K$ the counts are $c_{++} = \lfloor x + \frac{1}{2} \rfloor$ and $c_{-+} = b_3 - N_\phi/2$ with $b_3 = \lfloor (2N_\phi(C_K - R(+, +)) + C_K)/(2C_K) \rfloor = N_\phi + \lfloor \frac{1}{2} - x \rfloor$. Hence $c_{++} + c_{-+} = N_\phi/2 + \lfloor y \rfloor + \lfloor 1 - y \rfloor$ with $y = x + \frac{1}{2}$, which is $N_\phi/2$ when $y \notin \mathbb{Z}$ and $N_\phi/2 + 1$ when $y \in \mathbb{Z}$. The tie condition is $2x \in 2\mathbb{Z} + 1$, that is $2N_\phi R(+, +)/C_K$ odd. The absence of ties at the listed N_ϕ is the computation over every setting pair [5]. $\square$

No-signalling in the arrival intervals and in the counts: the arrival interval at either arm is that arm’s flight alone, independent of the other arm’s setting and length (Definition 1, the flight blind) (that interval the host’s tick at the arm’s set, GAMEBOARD [6, record 768]; the arm’s own count read under the key `clock_stamp` in the registered pair worlds of the engine before 2026-09-23, history; every world of the GO stamps the click with the detector’s count); neither party’s recorded outcome exists as a row before the completion, whose interval is the last arm’s, so the availability of the record, unlike the arrival, waits on the far arm’s length, and the paper claims no independence for that latency; the first party’s outcome is determined by u at the record’s birth (Theorem 11) and the second’s at the completion by the joint weights, both written at the completion, with the marginal $N_\phi/2$ for every setting (Theorem 11); so neither the counts nor the arrival times at either arm carry the other arm’s setting or length, and the one non-local step of the law (P6) is the assignment of the second party’s outcome from the joint weights at the completion, on the labels; its interval, the record’s, is the last arm’s. What the theorem does not cover, stated here. The wheel of the registered pair worlds runs at the rate 1 against N_ϕ , one birth per u in order, so the first party’s outcomes over the births are $N_\phi/2$ of $+$ followed by $N_\phi/2$ of $-$, a square wave whose serial correlation over a cycle is $1 - 4/N_\phi$ (0.9990 at 4096), a prediction of the sequence in time that no run reads and no row compares with nature’s record of outcomes, which shows no such order. The marginal is for settings fixed or chosen independently of the wheel: a setting chosen from u ($a = 0$ for $u < N_\phi/2$ and $a = N_\phi/2$ after, $b = 0$) gives the second party $+$ at every one of the 64 births at $N_\phi = 64$. On the register the settings are declared per world or balanced over every u (the choosers’ 960 births meet every u with every pair once), so nothing chooses a setting from u ; nature’s freedom-of-choice loophole has its counterpart here in the wheel, not closed by the register, and the reading that would test it is a marginal off $N_\phi/2$ with the settings chosen

from u [49]. And the order of the second party's outcomes carries the first party's settings even when they are independent of the wheel: the ladder's cells give $B = A$ at $a = b$ and $B = -A$ at $a = b + N_\phi/2$ exactly, so with $b = 0$ a first party keeping $a = 0$ leaves the second a serial correlation $\langle B_t B_{t+1} \rangle = 15/16$ over 192 births at $N_\phi = 64$, and one cycling $a = N_\phi/2, 0, 0$ (the period 3, coprime to N_ϕ , the independence criterion above) leaves $-5/16$, the marginal 96/96 in both, the same at every shift of the wheel and of the chooser (a computation from the theorem's cells, not a run): the second party reads one bit of the first's choice from the order of its own outcomes. So the theorem's no-signalling is of the counts over a cycle and of the arrival times; the sequence in time signals under the sequential wheel, a result of the law as built and a FAIL of any claim beyond the counts, open until the wheel is other than a counter (P4 allows no draw; the declared form below is a permutation from a world input, not a draw), or the claim is kept to the counts, as it is here. Stated plainly: under the counter wheel the law as built did not meet operational no-signalling in the order of its outcomes (its counts blind to the other party's setting, its order not), a fail by derivation; under the declared form it does, in form: the births of the pair record take the residues of $\mathbb{Z}_{N_\phi}$, one per u , in an order set by the world's seed through a declared integer hash (a Fisher–Yates permutation under a 64-bit SplitMix64 mixing hash) in place of the counter's $[1, 1]$; the click is stamped with the detector's count and the birth interval, never u ; the seed is an input of kind 1 drawn once per world, independent per world and of the settings, never read Outside; one permutation per pair lamp for the run, drawn once at load from the seed, the same for every cycle, the order repeating every W births (a fresh permutation per cycle is not the declared form; an independent draw per birth is what P4 forbids); the count channel under the declared form is item 21's statistic, the model's own prediction; the lamp's cadence is fixed and independent of the residues; the click's stamp is the cell's own first rung [6, records 1647 and 1648]. Stated honestly: the nonlocality is Inside and declared, one gather, the flips against the residues E_N a GameBoard diagnostic; the model's standing is Bohm's under quantum equilibrium [74, 75]; for settings held fixed over a wheel cycle the counts are no-signalling exactly and no reader Outside without the seed beats 1/2 on the order; for per-trial settings the marginal is no-signalling in distribution, and the order of the second party's outcomes carries the first party's switching pattern at order $E_N/(W - 1)$, the finite grain's own residue, stated as the model's prediction [22, item 21]. The joint weight J^2 is the click's evaluation of the one record through the two tables, a declared read-out (Definition 1; Theorem 10's characterization), which the paper names the one non-local step and does not derive from local transport; the group of order 24 is the lattice's rotations, the symmetry the rule is built on, and no result in the paper needs the 24 rather than the 48. Row 1c reads PASS in form under the declared form; its run confirms the counts and reads the flips as the diagnostic. The order channel's width at $b = 3N/8$ equals the finite- N correlation E_N exactly ($1448/2048 = 181/256 = S_N/4$), a closed form and not a statistic, a GameBoard diagnostic; the birth residue u is a hidden variable known at both stations, the birth stamp travelling with the record, so the counter form of the click is a deterministic hidden-variable model, and Bell's theorem forces its nonlocality into the order of outcomes, one way, from the first party to the second (the second's rung computed from $R(o_A, o_B)$ with a and o_A supplied), which nature's quantum mechanics does not carry: the counter form's own falsifiable departure from nature, declared before any run [15]. Read after a detector on the registered pair world of the engine before 2026-09-23, history (the full record [51, 52], row 1c): the second party's serial correlation 15/16 under a constant setting and $-1/16$ under the period-3 cycle 0, 21, 42 (the cycle $N_\phi/2, 0, 0$ the register's keys cannot declare; the declarable cycle reads the same channel with a smaller swing), the marginals 96/96 under both, the algebra met at 16 of 16 checks, nature's value 0: FAIL [42].

The strict-crossing rung $N_\phi C_k > u C_K$ would give the second party 33/64 in 3944 of the 4096 setting pairs at $N_\phi = 64$, a signalling of $1/N_\phi$ made by the rounding [3]; the nearest rung is what

the theorem needs. The theorem's scope is what it states: a pair of two labels with equal weights, the CHSH form, and the rung of (??); unequal weights and more than two labels are not covered, and no-signalling for them is open. That the marginal at B is $N_\phi/2$ for every a although B 's rung is computed from $R(o_A, o_B)$ with a and o_A supplied by the layer (Definition 1) is the content of the theorem; the joint counts are not local, and the model does not claim they are.

Theorem 12 (The finite- N_ϕ Bell value). *With the counts of Theorem 11 and off a tie, $E_{N_\phi}(a, b) = 4c_{++}/N_\phi - 1$ and*

$$|E_{N_\phi}(a, b) - \cos(2\pi(a - b)/N_\phi)| \leq \frac{2}{N_\phi} + 4 \arcsin \frac{\sqrt{2}}{2\varrho} < \frac{2}{N_\phi} + 0.0111, \quad \varrho = N_t - \frac{\sqrt{2}}{2}, \quad (16)$$

so at the labels $(a, b, a', b') = (0, N_\phi/8, N_\phi/4, 3N_\phi/8)$ (this assignment and its reverse the one quadruple of the four labels that gives 181/64; $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$), $|S(N_\phi) - 2\sqrt{2}| \leq 8/N_\phi + 0.0444$. $S(N_\phi)$ is an exact rational for each N_ϕ ; it lies above $2\sqrt{2}$ for 252 of the 512 multiples of 8 up to 4096 (among them $N_\phi = 16, 32$ with $S = 3$ and $N_\phi = 128$ with $S = 23/8$) and equals 181/64 = 2.828125 at every power of two from 512 through 8192 and $5793/2048 = 2.828613$ at 16384 [5]; measured after a detector at 64 and at 512, 1024, 2048, 4096, 8192 and 16384 [4] (the $S(N_\phi)$ figure of the engine before 2026-09-23, history [52]).

Proof. $c_{++} = \lfloor x + \frac{1}{2} \rfloor$ with $x = N_\phi R(+, +)/C_K$ gives $|c_{++} - x| \leq \frac{1}{2}$ and $E_{N_\phi} = 4c_{++}/N_\phi - 1$ from the mirror symmetry, so $|E_{N_\phi} - (4R(+, +)/C_K - 1)| \leq 2/N_\phi$. The vector $(C'[s], S'[s])$ differs from $256(\cos \theta_s, \sin \theta_s)$, $\theta_s = \pi s/N_\phi$, by at most $\frac{1}{2}$ per component, so its direction differs from θ_s by at most $\delta = \arcsin(\sqrt{2}/(2\varrho))$ with $\varrho = N_t - \sqrt{2}/2 = 255.29$, below every table vector's length (each component within $\frac{1}{2}$ of $256 \cos$ or $256 \sin$), so $4\delta = 0.01108 < 0.0111$ and $16\delta = 0.04432 < 0.0444$; $J(+, +)/\sqrt{n_a n_b}$ is the cosine of the angle between the two directions, $\cos(\theta_a - \theta_b + \epsilon)$ with $|\epsilon| \leq 2\delta$, and $4R(+, +)/C_K - 1 = 2 \cos^2(\cdot) - 1 = \cos(2(\theta_a - \theta_b) + 2\epsilon)$, within 4δ of $\cos(2\pi(a - b)/N_\phi)$. The four settings add the bounds. The values of $S(N_\phi)$ are the computation from these formulas with the repository's tables [5], which reproduces the design's 176/64, 720/256 and 2896/1024 at $N_\phi = 64, 256, 1024$ and the engine's measured values below (the paragraph on the finite-grain value). $\square$

What Theorem 12 does not say: that $S(N_\phi) \leq 2\sqrt{2}$, false for about half the N_ϕ ; nor that the deficit is $O(1/N_\phi)$ from below. The tables' term in (16) is the resolution $1/256$ of the circle's tables and does not shrink with N_ϕ ; at the CHSH labels the tables' entries are the same at every N_ϕ , so the rounding gives 181/64 at the powers of two from 512 through 8192 and $5793/2048 = 2.828613$ at 16384 and 32768 (measured at 16384, DETECTOR), and beyond the tables' bound $2N_\phi \leq 65536$ the rungs on the fixed correlations are the closed form $S(N_\phi) = 8(c_1 + c'_1)/N_\phi - 4$ with c_1, c'_1 the ladder's first cell's counts, whose limit with the tables fixed is $186034/65773 = 2.828425$, 2.1×10^{-6} below $2\sqrt{2}$; $2\sqrt{2}$ is the joint limit in N_ϕ and N_t [5, 4]. At a tie $c_{--} = c_{++} - 1$ and E_{N_ϕ} drops by $2/N_\phi$; no tie occurs at the CHSH labels up to 4096.

The finite-grain value, and what would make it a prediction. The construction's number is

$$S_{U24} = \frac{181}{64} = 2.828125 \text{ exactly,} \quad \Delta S = S_{U24} - 2\sqrt{2} = -3.02 \times 10^{-4}, \quad (17)$$

at every power of two from 512 through 8192, with the tables at $N_t = 256$, the wheel at $N_u = N_\phi$, the settings $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$ and a pair of two labels with equal weights; the marginals exactly $1/2$; the correlations at the two pairs of settings 725/1024 and 723/1024 at $N_\phi = 4096$,

above and below $1/\sqrt{2}$; 5793/2048 at 16384 and 32768. It was computed in advance and measured after a detector at $N_\phi = 512, 1024, 2048, 4096, 8192$ and, the plateau's end, 16384 (the last three [4, at b34fe114]; the $S(N_\phi)$ figure of the engine before 2026-09-23, history [52]), under the click and the wheel with every pin met (the register's block of the derivation's 24.4); 512 and 4096 were re-run by the second runner [7]. The criterion: the law at a grain is rejected when a measured S differs from $S(N_\phi)$ by more than five standard errors, under the identification of the experiment's settings and detection with the labels and the ideal read-out, which the model does not model; the comparison with nature is Section 8.1. It is a prediction of the law only once the pair's wheel meets no-signalling in the order of the outcomes (above), which the sequential wheel as built does not; until then it is the exact consequence of the declared construction at its grain.

6 The GameBoard: the way the algebra is computed with

The GameBoard is the way the algebra is computed with, not the object: a system of information transfer with costs, read in two ways, the detector's click a measurement and the host's view a diagnostic. This section states the law as one map with its postulates, the state it acts on under the record kinds of Sections 2 and 3, the six operations as the engine runs them, the read-out, the code and the ledger; every sentence is the algebra document's [9, chapters 2, 3 and 8]. The law as the engine ran it before 2026-09-23, the rows that hop whole with their wheel, their rungs and their components, is its record [52].

6.1 The law and its implementation

The one map. The law is the map of Eq. (??) applied at every Node at every interval, per component s of the state vector, each with its declared rate r and wall d : the component is translated by its rate, counted in walls, capped at $a = 1$ where a cap is declared and uncapped elsewhere (the whole part $\lfloor |s|/d \rfloor$ with the sign), and reduced by the walls it holds; every wall counted is an event, and nothing else happens. One declared variant of the count, the centred step, takes the nearest whole number the accumulator holds, $(|s| + \lfloor d/2 \rfloor)$ divided by d with the sign; it is named where a rule uses it. Where the rate is a constant of the world (the linear block) every component has the closed form $s(t) = \lfloor s_0 + rt \rfloor$, r the rate as a fraction of the wall, so a record's state at any interval is a formula of its birth and its age; where the rate is a function of what arrives (the feedback block: the push, the drive, the coupling of the two kinds) the map is iterated and its continuum limit is the differential equation $ds/dt = \mathbf{r}(\mathbf{s})$. The read-out is one comparison, the click [9, chapter 2].

Rules, parameters, initial conditions. The law is stated as rules, apart from the parameters and the initial conditions a world supplies. The rules: (P1) every quantity of the state is a bounded integer and every operation integer arithmetic, no real number, root or floating point at run time (the boundedness a property of the declared worlds over their declared ticks, guarded by the run-time amplitude bound: the light pair puts the pace at the six-neighbour rule's causal limit, $c^2 = 1/3$, so on a full three-dimensional board the two modes $\mathbf{k} = 0$, a uniform level, and (π, π, π) , the checkerboard, are marginal, double roots, growing at most linearly if inserted, and no GO world inserts either, and what a sharp edge excites of them on a full board the run-time bound watches; on a chain or a layer the band edge lies inside the band); (P2) space is a cubic array of Nodes, each joined to six neighbours by Links through directional Ports, a Node's complete local information its NodeState, time a sequence of intervals; (P3) the rates and walls of the map are made of the six operations of Section 1 and nothing else; (P4) a rule reads its own record and the six neighbours; nothing is kept at a Node beyond the events there, no register or draw, and the remainder r of

Eq. (9) is a component of the record's own state (the pair travels with the record [9, 8.8]), not a register at the Node; (P5) no row or body crosses more than one Link per interval; (P6) a record is read out once, by one comparison, at the click, the one step that deletes, and the click of a pair is one gather of one record from both settings, the law's one non-local operation. Beside P6, once: the pair is one record with two arms (Definition 1, the record $\psi = (00) + (11)$ of rank 2, each arm carrying the weights, nothing of one arm written on the other); the completion is one gather at the completion interval, no interval crosses between the arms and no arm's history is altered; the counts' no-signalling follows from the table's symmetry (Theorem 11); the completion is an operation of the law that relaxes P4 at that one step only, said so and not hidden. The parameters (P7, P8): the circle N_ϕ and its tables at $1/256$, the pair [num, den] of each record kind with its pitch, light the value $[1, 1]$ (Eq. (9)), the rung W of each detector, the cells of each block with its lowered pair, the take of each absorbing object, and the roundings declared at load; the initial conditions: the box and its boundaries, the blocks with their momenta, the lamps and the detectors of the world file, and the seed, the bound mode's integer profile over the whole board, the engine reading integers [9, 8.7]. The law of this paper is the runtime at the commit Appendix B names. The rules chosen among few (P9) are assumptions, not derivations, each named where it enters and each marked in the algebra document as carried, proved or computed and not proved: the rule with the pair per kind, carried from the design, its band and its two paces proved; the click's definition at the rung, carried on the author's word; the one formula of a block's motion, carried from the continuum, its two limits proved; the stress of light at a block's outer Ports as the wave's momentum flux, the design's declaration; and the bijection of the interval, the design's proof carried (Section 3). One axiom of the apparatus (P10): the click stores one pointer per set, the least-rank member of the family Theorem 10 forces, which is Born's form with the fundamental harmonic; which harmonic is a relabelling of the declared rate, and the harmonic constants are not derived. One assumption of the frame (P11): the GameBoard, Inside, is read only through its emitters and detectors, Outside; every result Outside is a click, a number of the board's state is never a reading, and every comparison with experiment is a comparison of clicks with readings; a measurement is itself an event of the board, an emitter putting a record on it and a detector ending the record at the click, the one step that deletes Inside and the one row that Outside gains. Nothing outside this list is assumed.

The state. The GameBoard is a box of Nodes in $\mathbb{Z}^3$, each joined to six neighbours by Links through six Ports, an axis periodic or open; time is the interval $t \in \mathbb{Z}$; the state at an interval is a vector $\mathbf{s}$ of bounded integers on a product of circles and lines. A Node's complete local information is its NodeState: per record kind present, the record's row, its two levels and its remainder ($a_{\text{now}}, a_{\text{before}}, r$) of Eq. (9); at a block's cell, the block's lowered pair, its pointer in the units of the conserved form and its count; nothing else, no register, no draw, no field beyond the records there; beside the click's gather the model declares two other non-local objects, a body's cells acting as one (the foreign object's extent, an input of kind 3, a detector of itself across its own cells) and the record's own ledger (its pointer per cell, its u , its norm), the record's state travelling with it and not a Node's, so that a Node holds nothing but the rows present at it and the locality claim stops at the Inside's verbs [9, 2.11, 8.1, 8.6]. A block carries its cells, its momentum $\mathbf{p}$ in $\mathbb{Z}^3$ and its step's accumulator per axis with its remainder (Section 3). This paper states one law, the record kind; the rows that hop are its record. The rows that hop, a tuple (Node, direction, age, phase, emitter, content, record, label, amount, multiplicity) moving one Link per interval on a digital line, are the engine as built before 2026-09-23, kept as their record [9, 2.11]: Definition ?? and the theorems stated on them (Theorems 8, 9, 10, 11 and 12) stand as proved, on the objects they name.

The six operations as the engine runs them. The order of the six verbs within one interval, one line each in symbols [9, 2.1 to 2.6, 2.11], and what is not one of them (a root of the state at run time, a float, a true division, a draw) are in the records file [52].

The three tests, rule by rule. A rule enters the law only if it is generic (one primitive with declared integers, no family name or kind, the engine branching on no name), vector (one of the six verbs on the state vector, its rate at most bilinear in the state, no root, no float, no rounding at run time beyond the ones declared at load) and local (its own record and the six neighbouring Nodes, fixed work and storage for a fixed K , nothing kept at a Node); generic means commuting with the 48 [9, 2.8]. Each rule of Section 3 passes the three and carries its mark [9, 8.10]: the rule with the pair on the six reads, (B), (T) and (D), the band and the two paces proved, the range-one bound computed and not proved; the conserved form, a bilinear form of the two levels, proved on every board with the remainders' identity; the well of the pair on declared cells, the threshold and the tail computed and not proved, the norm bound proved for the declared pairs whose band lies strictly inside the stability limit ([800, 809] bounded); under $[1, 1]$ the $k = 0$ and (π, π, π) modes are marginal and grow linearly, neither inserted by a declared world, P1's boundedness per declared world; the motion, the one formula carried, its two limits proved; the coupling, (B) linear in the other kind's two columns and one (D) per row, the Euler–Lagrange scheme and the invariant J with its remainder identity proved, the index and the pump computed and not proved; the click across the block's cells at W , (E), the bilinear form and (D)'s comparison, the definition carried; the take, a declaration, no identity; the seed, the standing start proved; the direction of time, (D) with the ceiling, the bijection carried and its inverse exact; the block's momentum, step and push, the cadence and the bound proved as identities of the accumulator, the stress carried as the design's declaration; the force between two blocks through light, the closed form carried, on the GameBoard computed and not proved.

The read-out. The click is the evaluation (E) of a body's own record on its cells at the declared rung W : light arriving at the block's cells drives the block's record through the coupling, the block's pointer accumulates its own record's motion across all its cells in the units of the conserved form, and the click is the comparison $W \times (\text{the pointer}) \geq (\text{the record's norm})$, integers both, in the block's own clock (its own record's cycles across its cells, not a count of clicks); the click line carries the block's own count and the light record's birth stamp. Light clicks only at a detector set, a declared set of Nodes with its rung W ; at a massive body's cells light does not click but drives the body's record through the coupling, and the body's own click reads that; a free Node has no rung, no count and no reading; the verbs are lossless, light passes on, and the click needs no sink; where a world must absorb it declares a take, a damping pair on light's row at the cells of an object declared absorbing, a declaration of the world and not the law's [9, 8.6]. A click is an action, not a passive read: the record ends at the detector, its content enters the detector's own record, and the detector's count advances, the click stamped with that count [9, 3.1]. Between a birth and a click the interval is a bijection, its inverse exact (Section 3, the direction of time; for the rows that hop, Theorem 9); the click is the one deletion, and the pair's gather the one non-local step (P6).

The code as the law's runtime. Every registered world is run from its world file with its pin written before the run and its source fingerprint (Appendix B); the books balance at every interval in every run; the design's scripts that gate the code, each printing its record beside the design and none an engine run, are listed in the records file [52].

The ledger. The forcing ledger, in the paper's records file [52], is the paper's spine: for every result, the rules of Eq. (??) it starts from, the assumptions added beside them with their kind, the

freedom left, and the ground of the derivation [34, 2026-09-22]: exact on the GameBoard (no limit, no average), a limit of the grain taken on paper with a rate, or a limit under a spatial condition, an average that covers the shell (over the shell's Nodes, over the directions of a fan declared in a ball, or over the sweep of an orbit), the condition then part of the claim. A result whose only reading is of the GameBoard's state is not in the table, and a result resting on a hypothesis beside the law or conjectured from the algebra stands in it only with that ground named in its row (the bending's coefficient, conjectured); the hypotheses beside the law are on the tree (Section 4).

7 How the algebra was reached: from the GameBoard to the group

The order of this paper is not the order of the finding. The object of Section 2 was reached from the GameBoard: from one Node and its six neighbours, one message per Link per interval, through the simulator's runs, in dated steps; what did not converge to the group and the ring was not put in. This section states the one choice and what it forces (the algebra document's chapter 7 [9]), the road from the operations to the group, the dated steps, the platform on which a formula is arrived at by two roads, and the check of the roads; the two direction tables, the simulator's part and the dictionary of the rows that hop, as the paper carried them, are its record [52].

The one choice, and the seven Nodes. The one choice of the model is the six Ports of a Node, $P = \{\pm x, \pm y, \pm z\}$ with the opposite involution, the L^1 neighbourhood of the cubic lattice: a Node and its six neighbours, the seven Nodes of the causal front, one message per Link per interval in each direction (Section 2). Two more choices, and no fourth: the phase circle $\mathbb{Z}_{N_\phi}$, N_ϕ per world, and the translation group of the torus, $\mathbb{Z}_X \times \mathbb{Z}_Y \times \mathbb{Z}_Z$, each factor a circle or a segment; the point group is then forced and is already the largest, the translation group is the board, and the only freedom that changes the physics is the neighbourhood, through the front, c and the tables [9, chapter 7].

What the choice forces: the 48, and the 24 within them. What is not chosen is the group of 48: every map that preserves the six Ports as a set and the lattice's Links is one of the signed permutations of the three axes, $3! \times 2^3 = 48$, and every one of them does; no three-dimensional lattice has a point group of order above 48 [9, chapter 7], and the cubic lattice's is the full octahedral group of that order, so this lattice keeps the 48 and no more. Nothing was added to it: the mass, the charge and the other contents of the family table live on the amounts, on which the 48 act trivially. The only $\mathbb{Z}$ -linear bijections that preserve the cone of one Link per interval are these 48; no boost preserves the polyhedral cone $|\mathbf{x}|_1 \leq t$ (whose linear automorphisms are finite), so no boost is among them, and Lorentz's form is reached Outside from the clicks (Section 4) and not as a symmetry of the lattice. The determinant splits the 48: the 24 rotations, the group of order 24, isomorphic to S_4 on the cube's four body diagonals, and the 24 reflections; the hand, needed to carry spin and polarisation, is the pseudoscalar that tells them apart, $h \mapsto \det(g) h$ (Theorem 1); the 24 are the octahedron's rotations, not a numeral chosen. The GameBoard showed that everything converges to this group and this ring, and what did not converge was not put in (the author's sentence, record 1109 [6]).

From the operations to the group. The operations did not deduce a group; they came first, as acts on the board: a row flies one Link per interval along its direction, rows meet at a Node, a body steps by an accumulator against a wall, a phase turns, identical rows merge, a detector clicks; each was then written as one of the six operations on a state vector. The group is the answer to which maps preserve what the operations act on, a Node of six Ports and its Links and a cone of one Link per interval: the bijections of the Ports that keep opposite Ports opposite are the 48 signed

permutations, and they are the only $\mathbb{Z}$ -linear bijections that preserve the cone, no boost preserving the polyhedral cone $|\mathbf{x}|_1 \leq t$ (whose linear automorphisms are finite), so Lorentz's form is reached Outside from the clicks and not as a symmetry of the lattice (Section 4). The 48 first appeared in the code as the invariance of one operation, the collision's table, and were recognised after as the symmetry of all of them; the 24 came from the hand, the pseudoscalar that carrying spin and polarisation needed, which $\det(g)$ keeps or flips. The other objects came the same way: the step and the flight are shifts, hence the translation group; the turn adds modulo N_ϕ , hence $\mathbb{Z}_{N_\phi}$, and the merge of records is the addition of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$; the collision is a permutation of a finite set; and c was not chosen, it is the flight operator's norm. What the group does back is to force forms: a rule that reads only the local state is required to commute with the 48 (the P-rule, a requirement beyond locality), and the symmetric integer matrices fixed by the 48 are the multiples of the identity, so the only isotropic quadratic reading of a body's momentum within the six operations is $\lambda \mathbf{p} \cdot \mathbf{p}$ with λ a constant, and the square W of Eq. (14) is assembled from three things derived separately, the 48 forcing the form's isotropy (every quadratic invariant a multiple of $\mathbf{p} \cdot \mathbf{p}$), the 3 the chain's $1/c^2 = 3$ and $E_0'^2$ the band's bottom; “generic” in the three tests of every rule means exactly this. The definitions and this road, with their code and tests, are the group-structure page of the tree [28]; this paper writes no second definition.

The transition from ordinary physics to the algebra, dated. The twenty-four dated steps, each a thing of physics becoming an object of the law with the reading that showed it, are [38] (`docs/designs/algebra_transition/HISTORY.md` on `main` at 412f5c61), their dates once: 2026-09-17 the shared resource deleted and the amplitude a phase with a content; 09-18 the Born table from N_ϕ and a Node its six Ports; 09-19 the quantum an integer row, the click the one border, the push one bilinear form, $E = hf$ a release's cost; 09-20 the readings by kind, the masses the initialisation, every count an accumulator, the fan a width of the law; 09-21 the six verbs, the record in $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$, the lattice the translation group, the Lorentz factor refused as a seventh verb; 09-22 the detector's clock, the click theorem and the framing.

The platform: how a formula is arrived at, by the algebra and by the simulator. Every formula of this paper is arrived at on one platform, by two roads that meet at the detector. The algebraic road has three steps. The Inside step: the six verbs on the record's attributes, the family's declared integers naming which attributes the step reads (Section 6.1; the families table [52]). The conversion: an emitter puts a record on the board and a detector ends it at a click, so that Outside there are counts and Nodes only, the detector's own count the age wall's member at coefficient 1 (P11). The Outside form: a ratio or a difference of those counts, the image of the Inside step, with its rung (exact, in the fan's limit, in the shell mean) and its order in the grain; a known form, Einstein's or Young's, stands on the comparison side only. The simulator's road runs the same three steps: a world file declares the families, the emitters and the detectors, and the reading is pinned before the run. Every reading of the simulator's road is DETECTOR in its count (each series with its detector in the conversion table [52]), the self-creations' ticks of series S GAMEBOARD; where a count is taken at an open-face detector without a body (the redshift rows 3, 4b and 12) it is read at head in the detector's own clock, its count of self-creations under the age wall ($r = 1.0000$, no crowd at the detector; [49], the re-run of 2026-09-22), equal to the host's interval to the integer, DETECTOR; the registered run's time base was the host's interval [6, record 768], superseded; a two-way light clock at the detector, not read, reads the same count in no crowd (series X); and a number of the board's state is a diagnostic, never a reading. A formula enters the paper with both roads: the algebra's chain with its inputs and its word (the roads table [52]), the simulator's reading with its series and kind (the conversion table [52] and the $S(N_\phi)$ figure of the engine before 2026-09-23,

history [52]), and nature on the comparison side (the full record [51, 52]). The platform is the paper's offer beyond its list: to arrive at a further formula, declare the family and the attributes it reads, write the Inside step on them, name the detector and what it counts, take the ratio Outside, pin it before the run, and read it by kind; the three chains below do this in full.

The road to each, and the check that nothing was derived from them. The three roads are the one chain of Section 4, each from the Inside step, the six verbs and the hypotheses named where they enter; the road to each formula and the check that no formula of nature was put in are in the records file [52].

8 The comparison with nature, and the failures by cause

Everything compared here is a detector's count or a ratio of counts (Section 4); the readings are compared by kind, and each failure is named with its cause.

8.1 The simulator's checks and the comparison with nature

Three things are kept apart: the gates on the code; what is shown Inside by algebra under named assumptions, which needs no experiment; and what is measured, a run Inside read Outside, compared with known experiments.

The formulas after a detector. Every number read after a detector is a count of clicks or a click's tick at a declared detector, registered with its source fingerprint (Appendix B; [4, 7]), its expectation written before the run; a number of the GameBoard's state appears nowhere. Seven formulas shown Inside (the pace, the click's power, Mach-Zehnder, the marginals, GHZ, Malus, light beside a mass) were confirmed by their runs on the engine before 2026-09-23, history (Appendix B); the readings that no formula gives are the conversion table's [52] (the clock's field at two Nodes, the equivalence after a detector, the covariant readings) and the $S(N_\phi)$ figure of the engine before 2026-09-23, history [52]'s (S at the same seven grains).

The method. An experiment is a world file: the GameBoard's extents, the blocks placed on it as the bodies of the experiment, the lamps and the detectors, every one a declared object of Section 3; a result is clicks, a count between clicks on a detector's own record or a ratio of such counts; the pin, the number the algebra gives, is written before the run and no pin moves after a reading. Every row of the paper's list passes three checks in order [26]: approved in the algebra, its number following from the one law before any run and closed on the closure page [12]; approved on the GameBoard without pins, an exploratory run reading beside the algebra's number within the band declared before the run, labelled exploratory and never a result; approved with pins, the pin written before the run [14, 25], one run on a recorded decision, the clicks read by the reader of record and the reading by kind inside the pin. A row is (K), a known-formula row whose algebraic identity is a textbook formula and whose engine reading is the control of the engine against the algebra, or (P), a prediction row, a number the algebra gives and no known formula gives; a prediction is not a pass. A reading that falls outside its pin is written as a miss, in this table, by kind; every number of the law is stated as matching nature or not, never as how nature is. Table 1 is the paper's list at this version: every pinned reading is a blank, since no row of the list has yet been read on the engine as it stands, and the blanks fill as each run is read and merged; the old comparison of the engine before 2026-09-23, its rows, its readings and its figures, is history in the records file [52], never carried into this table.

The limitations, stated before the comparison. The third status follows from the structure of the law as built and never from the outcome, and a referee’s test that every cell of the table must pass is that a row sits in the same class had it agreed. The law as built derives a moving clock’s rate, $1/\gamma_m$ at the massive kind’s cone (Section 3), and not the Lorentz group: the GameBoard is a medium with a rest frame, Lorentz’s symmetry belongs to the limit and to the clicks, no Michelson–Morley or Kennedy–Thorndike row is claimed, and the light clock in motion is a predicted FAIL as declared (row LCm); the beam law’s bodies that hop had $r = 1$ at every speed and nothing contracted, no relativistic dynamics on them, history; it has no term of the crowd’s history, so no observable that reads the growth of an emitter’s stretch with the flight time is predicted; and a declared input is not a prediction, so an observable whose number is a declared give (row 7a), a declared content (row 8c) or a declared coefficient (row 13’s γ_{PPN}) is not predicted, whatever the declared world reads. A fan of directions bounded in Manhattan length is not a fan of every direction (its angular density is the octahedron’s, the axes 2.8 times denser than the face diagonals); a reading that assumes the isotropic mean holds only where the fan is declared uniform in angle at a grain finer than the reading, or where a click read the crowd; otherwise the law’s own number for that world is the integral on the comb (the register’s finding under rows 6, 10, 13 and 14). The beam law’s weight rule for its rows 13 and 14 (γ_{PPN} , the declared space-curvature coefficient) is in the records file [52], history. The count: the paper’s list of Table 1 is closed in the algebra before any run, every pinned reading a blank until its run is read and merged, and none is read yet on the engine as it stands; the old count of the engine before 2026-09-23 is history [52]; the disagreeing rows are the law’s own results, and the paper claims no rule it has not shown. Two placements, row 7a to not predicted and row 2a to NOT COMPARED, were decided on 2026-09-23 after the kind audit [29]; nothing else moved. A row whose apparatus table is the declared law (the polariser’s and the splitter’s tables) is checked as an implementation of that table, and only a row whose evidence class is DERIVED_BLIND is offered as a prediction; the class column of Table 1 fills from the closure page and not from this paper’s judgement.

Table 1 The paper’s list, closed in the algebra before any run [12, 26]: one line per row with its three checks. Kind: (K) a known formula as the thing compared with, (P) a prediction of the algebra with no known formula. Column 1, the algebra’s number, a COMPUTATION declared before the run, with nature’s number beside it for a (K) row; column 2, the GameBoard without pins, an exploratory reading labelled so, or not yet; column 3, the pinned reading by kind, a blank until its run is read and merged, then DETECTOR (a click, a count between clicks) or the word miss. Sources for nature’s numbers as the register cites them [51]. Class: the evidence class of the row, one of DERIVED_BLIND, APPARATUS_INPUT, CALIBRATED_FROM_TARGET, IMPLEMENTATION_GATE and FALSIFIER_ONLY, blank where the closure page has not fixed it (row R2); only a DERIVED_BLIND row is offered as a prediction. Version: every row’s algebra number is the law of 2026-09-23, the massive record kind with the local detector law (on `main` at `a867ab22` [9]); no reading of the earlier runtime, the beam law before `3a9a7109`, is carried into this table, and wherever the text cites one it is marked history.

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
1a	Bell’s CHSH sum S of a pair, two polariser tables reading the record’s phase	K	the table’s $\cos^2$ law on a shared record summed at the four settings, $S = 181/64 = 2.828$ at the declared grain $N_\phi = 2048$, exact, 1.05 standard errors from Poh et al.: a comparison at a declared grain, not a test of the grain (the measurement excludes the grains whose $S(N_\phi)$ lies outside its band; the shortfall of a measurement from $2\sqrt{2}$ is its apparatus’s); nature 2.42 ± 0.20 [82] and 2.82759 ± 0.00051 [81], verified [16]	not yet: the phase-reading component and the tables	_____	APPARATUS_INPUT
1b	The CHSH sum under the phase-form window	K	2, a control	not yet	_____	APPARATUS_INPUT
1c	The order channel	K	the counts no-signalling exactly (row 1d). The order of outcomes: under the counter wheel $[1, N]$ a FAIL by derivation (the second party’s outcome sequence flips with the first party’s setting on 1448 of 2048 births at $b = 3N/8$, the flip fraction $E_N = S_N/4$ exactly, open by Bell’s theorem for any deterministic wheel [15]); under the declared form (the residues in a seed-set order, one permutation per pair lamp drawn once at load from the seed and repeating every W births, the click stamped without the residue [6, records 1647 and 1648]) PASS IN FORM Outside: no reader without the seed reads the first party’s setting from the second party’s list (exact for a uniformly random order, the seed-set permutation its integer form), the flips against the residues a GAMEBOARD diagnostic of the declared Inside nonlocality; the finite grain’s residue, the second party’s sequential agreement rate carrying the first party’s switching pattern at order $E_N/(W - 1)$, 3.5×10^{-4} at $N = 2048$, is the model’s own prediction (COMPUTATION, [22, item 21])	not yet (the declared form’s run)	_____	APPARATUS_INPUT
1d	No-signalling of the marginals	K	0 exactly by the table’s symmetry, each party’s + fraction $N/2$ of N at every setting, 1024 of 2048 on the declared world	not yet	_____	APPARATUS_INPUT

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
2a	The two slits, one quantum at a time, the visibility	K	at or above the train's coherence value, the band 0.01 (0.959, 0.958, 0.958 at 12, 16, 24 Links for 32 periods); no nature number, NOT COMPARED	partial: the slits' exploratory page	NOT COMPARED, the placement of the limitations paragraph	APPARATUS_INPUT
2b	The Mach-Zehnder visibility	K	1.00 – 0.02, the dark port 0 of 64 at 128 periods; nature 0.98 [83]	not yet: the pair's two arms and the splitter's table	_____	APPARATUS_INPUT
2c	Born's exponent, the three-opening sum	K	the exponent 2 exactly, the click a quadratic form; the sum 0 to the remainder's grain; nature 0.0064 ± 0.0119 [62]	not yet	_____	DERIVED_BLIND
9	Malus at 45 degrees and at 11.25, 28.125, 33.75 degrees	K	128 of 256 at 45 degrees, 0 crossed, 64 with a third between; the table's counts of 256 at the three settings; nature $\cos^2$	not yet on the engine as it stands	_____	APPARATUS_INPUT
10	The single opening's spread, two halves	K	(a) 0.842 at $w = 2\lambda$, $F = 0.16$, the Rayleigh–Sommerfeld sum on the board; (b) 0.886 in the far field; both COMPUTATION [12, row 10 and its script]	not yet	_____	DERIVED_BLIND
7	The atom's lines at the coupled modes	P	the lines at the modes' own frequencies, 0.0995 and 0.1432 radians per interval	exploratory, GAMEBOARD: the coupled mode 0.09948 within 0.3 percent	_____	DERIVED_BLIND
4a	The muon in flight, the moving clock at $k = 3$	K	0.8116 by the one formula at the exact cone on the declared layer (the pair [800, 809], $s = 14$), 0.8108 at c_m and 0.8132 at c beside as controls; nature's form $1/\gamma$ [60]	at rest: the well's mode period 42.364 against 42.36, no beat; in motion partial, the ramp redeclared	_____	DERIVED_BLIND
4b	The moving lamp's redshift	K	$1 + z = 1.9889$ on the declared world, the free limit 1.9339 beside, 1.9355 with the pair [800, 809]; nature's form $\gamma(1 + \beta)$ [61]	not yet: the redeclared geometry	_____	DERIVED_BLIND
4c	The round-trip Doppler off a receding transponder	K	$(1 + \beta_c)/(1 - \beta_c) = 3.732$ at $k = 3$ exactly, the hop's sidebands named beside; nature the two-way form of every radar	not yet	_____	DERIVED_BLIND
5a	The anisotropy of c by direction	K	the phase pace by direction ($k^2/36$, Eq. (12)) within 0.8, 0.4, 0.2 percent of c at 12, 16, 24 Links, falling as the inverse square of the wavelength, GAMEBOARD; a PARAMETER BOUND, not an agreement: nature below 10^{-17} [58], 10^{-18} beside [59], the bound on the free grain [15]	not yet on the engine as it stands	n/a, a bound, no pinned entry	DERIVED_BLIND
R2	The Sagnac ratio on two pushed blocks	K	$\beta = v_c/c = 0.5774$ at $k = 3$ exactly, a count between clicks independent of the clock's factor; nature's fringe shift 0.230 ± 0.005 against the computed 0.236 [23] (the two inputs verified in published excerpts, [16, row 80]), the ratio 0.975 ± 0.021 ours, a CONVERSION; the clicks declared at $W = 256$: 108 at rest, 247 chasing, 72 meeting, each ± 2 ; raw $175/319 = 0.5486$, 0.5774 after each direction's own rise, the identity (DECLARATION, sections 13 and 15 at 25a7abf4 [14]; no reading yet)	not yet	_____	_____

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
LC	The light clock: one emitting block A and a mirror	K, P	the chain's closed face at $x = 672$ sixty Links from A's face; $N_0 = 213 \pm 1$ in A's own clicks at $W = 64$ on its returned record at the receiving set bound to A by name, the free Node at $x = 612$ (the set's first rung at age 213 by the derivation under the click line and the receiver by name, the geometry alone, the declarations' section 10, item 9, merged at ecebf895 [14]; the earlier form's 218 ± 2 , derived for the set at A's twelve cells, no longer in the world, kept as history), a pin declared before any pin run, $2L/c = 207.85$ beside as the (K) form, the rung's rise of about ten intervals the click's own; in motion (P) as declared, $\gamma^2/\gamma_m = 1.2224$ along and $\gamma/\gamma_m = 0.9981$ across (item 7), the light-held pair 1.0136 (item 8); "two bodies" and "two ring-ups" history	not yet	_____	DERIVED_BLIND
LCm	The light clock in motion, Michelson–Morley's null	P	the pushed pair at a declared separation: $\gamma^2/\gamma_m = 1.2224$ along and $\gamma/\gamma_m = 0.9981$ across at $k = 3$ by the band, COMPUTATION [13]; Einstein's 1 and 1 need the separation to contract, so the along arm is a predicted FAIL as declared (a rigid arm's anisotropy $\beta^2/2$, 5×10^{-9} at the Earth's orbital speed, against the cavity bounds at 10^{-17} [58, 59]) and the across arm is met in form, the residue $c_{\text{eff}}/c = 0.99627$ beside; nature's null [58] the falsifier; a light-held pair on the band (the declared world $\lambda_0 = 32.1$ Links at $k = 3$) reads $L_{\text{along}}/L_0 = 0.8127$ against $1/\gamma = 0.8165$ and the along ratio 1.0136 inside the declared band 1 ± 0.03 , COMPUTATION [18], the excess 1.36 percent 42 times $(a/\lambda)^2\beta^2 = 3.24 \times 10^{-4}$ on the declared world (COMPUTATION), tending to $(a/\lambda)^2\beta^2$ only in the limit, and the pair's settling at the nodes the named open step	not yet	_____	DERIVED_BLIND (P)
v	The index of a medium block at rest	K	$n^2 = 1 + Gg/(\omega_0^2 - \omega^2)$ in closed form, a control of the coupling; the cavity's mode sum beside it, the 4 percent between them ours, the cavity's mode sum against the closed form, open in size	exploratory at rest: $n - 1$ read at 0.87 of the closed form at all three couplings, a GAMEBOARD probe [19, the index block's row]	_____	DERIVED_BLIND
v-m	The index in motion	P	the scheme's own number on the declared geometry; Fizeau's form the declared non-match	partial, GAMEBOARD: head-on 0.498 beside	_____	DERIVED_BLIND
ii	The moving block's clock on 64^3 , two worlds	P	0.7814 and 0.8032 (clock ratios, dimensionless), the controls beside	exploratory: 0.7833 and 0.8055 by the peak, the three gammas indistinguishable at this board	_____	DERIVED_BLIND
M1	Matter waves through two openings	K	the side lobe's count centroid at $y = 64 + 27.79$ for $\lambda_{\text{dB}} = 12$ Links, the two-source sum with the band's k , the band one Node; nature's form de Broglie's	not yet	_____	DERIVED_BLIND

Row	Experiment	Kind	1. The algebra (COMPUTATION)	2. The GameBoard, no pins	3. Pinned (by kind)	Class
M2	The energy of a moving mass, $E = mc^2$ its rest limit	K	the first click 169 ± 2 intervals after the lamp's birth at the declared rung, the group transit 167.1 beside as the form; $m = k/v_g = 1.0417$ the band's number in the form of a moving mass	not yet		DERIVED_BLIND
A	The two-pace bound on the one scale	P	the deficit $\omega_0^2/4$ of c_m below c ; with the electron's c coefficient 2×10^{-14} [20] one interval is at most 3.6×10^{-28} s, with the one-direction Crab bound at most 6.3×10^{-31} s; a bound, no run	n/a	n/a, a bound	DERIVED_BLIND
A2	Light against itself, the quadratic dispersion by wavelength and direction	P	the coefficient $3 \sum_i n_i^4 - 1$ in $[0, 2]$ on the sky, Eq. (12), a prediction of the form; the bound on the one scale from the burst bound on quadratic photon dispersion, about 2×10^{-35} s, the published number verified against its source [16], the factor $3/2$ the paper's own reading, the sensitivity $\sqrt{2/3}$ beside [27]	n/a	n/a, a form and a bound	DERIVED_BLIND (the form); FALSIFI- FIER_ONLY (the bound, its source verified)
B	The bound clock's second term	P	$\epsilon(\gamma_m^2 - 1)/2$ below $1/\gamma_m$, printed on every massive world; a bound, no run	n/a	n/a, a bound	DERIVED_BLIND

The derivations, final. Four numbers of the list are the algebra's identities and stand whatever the runs read: Born's exponent 2 exactly, the click a quadratic form (row 2c); Tsirelson's value $2\sqrt{2}$ as the limit of the finite-grain Bell value, the local bound beside it (row 1a); $c^2 = 1/3$, the pace of light as the norm of the flight operator, one Link per interval on every line and $1/\sqrt{3}$ in the Euclidean mean (row 5a); and the rest energy of a massive kind, $h\omega_0$ with $\omega_0 = mc^2$ in the law's units at the massive cone, de Broglie's internal clock as the thing compared with (row M2).

Under the hypothesis, after the pins, on a recorded decision. Three rows enter only under the pair-field hypothesis, a third record kind sourced by every block's content and read by every Node as its own pair, outside the law under its own identity and never in the list's place: the clock in a field at r and $2r$, the ratio 2.00 (row 12, Newton's potential from the lattice Green's function); the delay of a light pulse through a medium block sitting in the field, Shapiro's form with the same $1 + \gamma$, one declared input (in place of row 13's bending); and Newton's fall, a prediction of the form and not a pinned row (in place of row 14's orbit).

The rows that left the list. Under the law as it stands the following rows of the register are not predicted, each with its missing piece named on the schedule [25], and their earlier readings are history in the records file [52]: the far lamp (rows 3 and 11a to 11c, a crowd's field from the rule), the nuclear bindings (7a and 7b, a binding energy from the rule), the decays and the neutrino (8a to 8c, a lifetime and a mass ratio from the rule), the two arms in motion (5b, kept as the prediction page of the light-bound arms), complementarity and aberration (R5 and R6, their declarations), Bohr's ladder (6, replaced by the atom's lines), the bending (13, replaced by the delay under the hypothesis) and the period ratio (14, replaced by the fall under the hypothesis). No row is removed from the register; a row not predicted says so.

The prediction rows. The (P) rows of Table 1 are the algebra's own numbers with no known formula behind them: the atom's lines at the coupled modes, the index in motion, the moving block's clock on the two 64^3 worlds, the ring-ups of the light clock, the two-pace bound, light's anisotropic dispersion on the sky and the bound clock's second term; each has its falsifier written before any run [9, 8.9], and a prediction is not a pass. The rows read on the engine before 2026-09-23,

with their agreements and their disagreements, are history [52]; what this version claims is the algebra's numbers before the runs, and the blanks of the table say so.

8.2 Discussion and conclusions

The chain from the click to Lorentz's factors, Einstein's step and Newton's form is Section 4; the two roads and their check are Section 7; what is proved opens Section 5.

What the law names and has not computed. Each item below is a computation the law's rules define, on a declared world with a declared click, that this paper has not made: the register holds no reading for it, nothing is claimed for it, and its verdict is NOT YET, NOT COMPARED or absent until a run is made and read by kind. (i) The lines of the atom: not a run not yet made but a rule the law lacks, no rule of the six quantising a body's loop and one click reaching only the loops with $2i^2 \geq j^2$ (the full record [51, 52], row 6; [46]); Bohr's 27/20 waits on that rule, under its own identity; the ladder itself, $a_j/a_i = (j/i)^2$ and $E_j \propto 1/j^2$ with 27/20, is conjectured from the algebra, not computed (the atom's item in the records file [52]). (ii) Third-order interference, Sorkin's κ_S on a three-opening world (row 2c; no such world is registered). (iii) The single opening's spread against its pin (row 10), and the rows pinned without their run (4a, 5b, 11). (iv) The quarks and the nucleon: the quark families are in the catalog with their rational charges and their content per record (the families table [52], u), and no detector reading of a quarks world is registered; the nucleon as three measured events bound by the strong column, its read mass against the sum of its parts, is the reading that would test them [6, record 248]; colour, a $\mathbb{Z}_3$ label that passes the three tests and does nothing beside a colour force between rows that fails the vector and the local test, is designed under its own identity and not built [6, record 270]. (v) The unseen content: on the law's own rules the field at a galaxy's edge is Newton's in the shell mean and the detector adds nothing there; the one route to a flat rotation curve without unseen content, a short periodic dimension ([31], entry 6), fails nature's Tully-Fisher slope on a host map ([48], no run); what the law admits is unseen content as an input family, as nature's model does; no rotation curve is read after a detector and no disc under a declared unseen family is run. (vi) The size of the crowd's stretch, $g_1 = 0.36$, that the supernova diagram needs, from a declared content over a flight (an input in the dark energy item of the list above; the crowd's own accumulation gives the wrong sign), and the periodic GameBoard against the open one on the measured numbers, declared and not decided [6, record 270]. (vii) The detector's constants c_j of the click's harmonics, not reached by the algebra (Section 5.4). (viii) The inverse square in space, pending its run (row 14); the bending's full deflection, its coefficient $2(1 + \gamma)$ on the one constant conjectured from the algebra; the ring's rule read inside its pins and the constant 4 not closing within the grain (row 13). Apart from these stand the numbers for which the law has no formula and this paper offers none: the hierarchy ρ_p , the constant $h = h_q N_\phi$, the families' content per unit and the grain N_ϕ , inputs by the dictionary (the forcing ledger [52]).

Positioning. The click's assignment of the second outcome from the joint weights is a non-local step on the labels, its interval the last arm's (as after Theorem 11), forced by Bell's theorem because the marginals are exact and $S > 2$ [53, 54]; no-signalling of the counts is Theorem 11 [55], and the order of the outcomes signals under the sequential wheel (Section 5.5; the full record [51, 52], row 1c, FAIL BY DERIVATION, the order channel computed before any run [15]). The click's finite kinematics is Weyl's and Schwinger's on $\mathbb{Z}_{N_\phi}$ [66, 67]; the transport a lattice gas's exact integer dynamics [56, 57], the flight table's transport free of dispersion where the lattice automata of [71, 72, 73] disperse, the record kind's not, its k^2 dispersion the prediction of row A2; the uniform

u does the work of quantum equilibrium [74, 75]; the model is ψ -ontic [76, 77]; Theorem 12 puts $S(N_\phi)$ on both sides of Tsirelson’s bound [79, 80]. The earlier testbed [1] is history.

Conclusion. One local integer rule on the cubic lattice, six operations required to commute with the group of order 48, is read only through detectors’ clicks. Exact on the lattice are the count marginals, the read-out’s quadratic form under its axiom and the CHSH sum as a rational of the grain; the moving clock’s rate is derived under the massive record kind; Lorentz’s factors and Einstein’s step are reached only under two named hypotheses. No pinned reading is in the table yet: every row is closed in the algebra before its run, and the pinned runs decide, row by row, what the law matches and where it fails.

Declarations

Funding. None.

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Ethics approval. Not applicable.

Consent to participate and consent for publication. Not applicable: no human participants and no animals.

Data availability. Every world file, every registered run’s record, the check scripts’ outputs and the derivation records cited in this paper are archived at Zenodo [84] (concept DOI 10.5281/zenodo.22738746); the commit of the archived version is named in Appendix B, and every number of the paper has one row in the ledger of that archive (`paper/general_formula/NUMBERS.md`) naming its kind and its source.

Materials availability. Not applicable: no physical materials.

Code availability. The simulator, the check scripts and the figures’ scripts are in the same archive [84], under its open licence; the figures are drawn by those scripts from the definitions, none by a generative tool.

Author contributions. Sole author: the guidelines of the project, the model, its rules, the worlds and the experiments, every reading and its verdict, and the decisions recorded in the archived log; the part of the large language model used as a tool is stated in the next declaration.

Use of artificial intelligence. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author’s direction and instructions, for the software of the simulator and its checks, the computations, the mathematical derivations and their verification, the physics of the design pages, and the drafting and editing of the text; the model is not an author and meets no criterion of authorship. Every guideline, postulate, hypothesis, rule, world and decision of the project is the author’s, recorded in the archived log; the author reviewed and edited every statement of this paper and takes full responsibility for its content. The author’s statement: I laid down every postulate and hypothesis of the model myself and brought in nothing external; the forms of Newton, Kepler, Einstein, Lorentz, Bohr and Balmer that this paper cites are the things compared with, not inputs to the model.

References

- [1] A. Gonen, A finite-integer causal-GameBoard testbed: which resources reproduce interference, causal collapse and Bell correlations on one GameBoard (2026), arXiv preprint.
- [2] The Beam Law and its implementation notes, `docs/BEAM.LAW.md` of the archived code [84].
- [3] The design of `amplitude-v1` and its check outputs, `docs/designs/amplitude-v1/` of the archived code [84].
- [4] The experiments register, `docs/EXPERIMENTS.md` of the archived code [84]; consolidated on the branch `register-paper-sources` (its head `b12dd253`, on `main` by PR 860).
- [5] `paper/general_formula/checks/s_of_n.py`, `s_powers_of_two.py` (the pair's S and its four correlations at the powers of two from 512 to 32768, the closed form on the fixed correlations to 2^{20} with the distances from [81], with the tables fixed), `which_path_window.py`, `tables_norm.py`, `information_transfer.py`, `limits.py` and `light_speed.py` with their outputs, `figures/summary.json` and `NUMBERS.md`, in the archived code [84].
- [6] The dated working log of the model, cited by the number of its record, `docs/LOG_2026-09-20.md` of the archived code [84].
- [7] The replications register, `docs/REPLICATIONS.md` of the archived code [84].
- [8] The families in the paper, the audit of the families table of 2026-09-21, `docs/designs/paper_families/FAMILY_TABLE_AUDIT.md` with its host script's output `family_table.out`, of the archived code [84] at the commit named in Reproducibility.
- [9] The algebra of the law, in one place, `docs/ALGEBRA.md` of the archived code [84]: chapter 8, the rule with a pair per family, the conserved form, the coupling and the direction of time; chapter 9, the tools as operations of the group, the one element and the one operator (9.19), the property test (9.20), the implications of one algebra (9.21), the family and the experiment (9.22), the two-qubit computer (9.23), the moving body (9.24) and the click over a set of cells (9.25); on the branch `lab-tools-unification` at `4b365063`.
- [10] The lab tools: one specification per tool (part A) and the experiments as placed tools with their pins written blind (part B), `docs/designs/lab_tools/LAB_TOOLS.md` of the archived code [84], on the branch `lab-tools-unification` at `4b365063`.
- [11] The massive record kind, the design page: the pair per kind, the band, the block, the coupling, the click, the seed and the direction of time, `docs/designs/detector_law/MASSIVE_RECORD.md` of the archived code [84], on `main` at `74e7a862`.
- [12] The algebraic closure of every experiment: the rows that close on a small board in minutes, the rows replaced and the rows dropped, `docs/designs/detector_law/ALGEBRAIC_CLOSURE.md` of the archived code [84], on `main` at `0a9511b5` (with the provenance word of every row).
- [13] The light clock in motion, derivation (a), the along and the across arms of a pushed pair, `docs/designs/detector_law/light_clock_motion_pins.py` and `DECLARATIONS.md` section 10 item 7 of the archived code [84], on `main` at `db584eb7`.
- [14] The declarations of the pinned worlds, every pin by kind before its run, `docs/designs/detector_law/declarations/DECLARATIONS.md` of the archived code [84], on `main` at `ecbf895` (the GO's declarations; its section 14 the tables acting on the record's pair in the linear form).
- [15] The order channel derived before any run: the counts' no-signalling exact and the order's flip with the far setting on 1448 of 2048 births at $b = 3N/8$, `docs/designs/detector_law/order_channel_pins.py` with its output and `DECLARATIONS.md`'s clause of the archived code [84], on `main` at `9707c866`.
- [16] The published numbers verified against their sources, one line each, `docs/designs/detector_law/SOURCES_VERIFIED.md` of the archived code [84], on `main` at `2aeea22c` (its rows 55 and 80 at `6f51f630`).
- [17] The mathematical verification of this paper, one line per statement, `docs/designs/paper_verification/MATH_READ.md` of the archived code [84], on `main` at `835556b7`.
- [18] The light-held pair on the band, the declared world and its along ratio, `docs/designs/detector_law/light_held_pair_pins.py` with its output and `DECLARATIONS.md`'s clause of the archived code [84], on `main` at `a867ab22`.
- [19] The detector law's exploratory readings, `GAMEBOARD` probes without pins, `docs/designs/detector_law/EXPLORATION.md` of the archived code [84], on `main` at `a5e80449`.
- [20] B. Altschul, Phys. Rev. Lett. 96, 201101 (2006), <https://doi.org/10.1103/PhysRevLett.96.201101>; Phys. Rev. D 74, 083003 (2006), <https://doi.org/10.1103/PhysRevD.74.083003>; the value and both articles verified, [16, row 55].

- [21] The pair as a field: the hypothesis `pair-field-v1` stated beside the law under its own identity, `docs/designs/detector.law/PAIR.FIELD.md` with its computation `pair.field.pins.py` ((a) the lattice Green’s function on 120^3 , (b) the clock in the field) and its printed output, of the archived code [84], on `main` at `d2d1d2fb`.
- [22] The order channel’s attacks page, the four attacks answered and the guard, `docs/designs/detector.law/ORDER.CHANNEL.ATTACKS.md`, with `ONE.ACCOUNT.md` (one account in four layers) and `DECLARATIONS.md` section 2, item 8 (the two keys), of the archived code [84], on `main` at `d2d1d2fb`; its item 21 (the finite grain’s residue, the model’s prediction) on `main` at `f94e8fad`, with the worst-case line $1/(W - 1)$ beside $E/(W - 1)$.
- [23] A. A. Michelson, H. G. Gale and F. Pearson, *Astrophys. J.* 61, 140 (1925), <https://doi.org/10.1086/142879>.
- [24] The closed glossary: one name per term, the retired names beside it and the source of each, `docs/GLOSSARY.md` of the archived code [84], on `main` at `0e479a22`.
- [25] The schedule of every row against nature under the new law: the world, the declaration, the pin by kind, the board, the host time, the day and the old reading it supersedes, `docs/designs/detector.law/SCHEDULE.md` of the archived code [84], on `main` at `53cec95d`.
- [26] The approvals file: one row per experiment of the paper’s list, three checks each, `docs/designs/detector.law/APPROVALS.md` of the archived code [84], on `main` at `054a6223`, beside the paper’s storyline, section 8 of `docs/designs/detector.law/PLAN.md`.
- [27] Light against itself: the GameBoard’s own dispersion of light, its anisotropy on the sky and the bound it puts on the one scale (row A2), `docs/designs/detector.law/LIGHT.DISPERSION.BOUND.md` of the archived code [84], on `main` at `1410d113`.
- [28] The group structure of the law: the four group objects, their operations, and how the group of order 24 and the 48 act on each, `docs/GROUP.STRUCTURE.md` of the archived code [84], on `main` at `7739bbbe` (the merge of the branch `group-structure`).
- [29] The kind audit of the register’s rows: every number’s kind re-read against its source, `docs/designs/fail.rows/KIND.AUDIT.md` of the archived code [84], on `main` at `46c27396`.
- [30] The six rows that disagree, with the number: why each disagrees and whether the experiment behind it was sound (2026-09-23; row 13’s suspect confirmed by one re-run at head, its section 9), `docs/designs/fail.rows/DISAGREES.md` of the archived code [84], on `main` at `103a30fa`.
- [31] Hypotheses under test, entry 27, the conditional derivations as declared hypotheses, `docs/HYPOTHESES.md` of the archived code [84].
- [32] Canonical simulation terminology, `docs/TERMINOLOGY.md` of the archived code [84].
- [33] The postulates of the program, `POSTULATES.md` of the archived code [84].
- [34] The project’s decisions and dictionary, `docs/HIGHLIGHTS.md` of the archived code [84].
- [35] The derivations of the beam law, `docs/DERIVATIONS.BEAM.md` of the archived code [84].
- [36] Light Outside from the board: every equation of light Inside carried Outside by its own transformation, `docs/designs/light.outside/DERIVATION.md` of the archived code [84], on `main` at `a62441fb`.
- [37] Einstein Outside: the formulas of the special and the general theory brought from inside the GameBoard to above it, `docs/designs/einstein.outside/DERIVATION.md` of the archived code [84], on `main` at `e8432e6c`.
- [38] The transition from ordinary physics to modern algebra: a dated, cited history, `docs/designs/algebra.transition/HISTORY.md` of the archived code [84], on `main` at `412f5c61`.
- [39] The click frame, part 2 (“no square is declared anywhere in the chain”), `docs/designs/click.frame/DERIVATION.md` of the archived code [84] on `main` at `af051aaf`, one file with the click frame.
- [40] New formulas of quanta and bodies: candidates that the Inside step gives Outside and the continuum cannot state, `docs/designs/new.formulas/QUANTA.md` of the archived code [84] on `main` at `477daa1d`.
- [41] New formulas of light: what the Inside step gives Outside that the continuum cannot state, `docs/designs/new.formulas/LIGHT.md` of the archived code [84] on `main` at `7370d524`.
- [42] The order channel: the run and its readings, `docs/designs/order.channel/RUN.md` and `PINS.md` of the archived code [84], on `main` at `804a8da0`.
- [43] The Gleason bound: the tables’ rounding against hypothesis (a), the weight of a mixture the register cannot exclude, and the table click’s zeros, `docs/designs/gleason.bound/BOUND.md` of the archived code [84], on `main` at `e6809713`.

- [44] The step algebra of light bending: the walk of a light row past a held mass by the law's integer steps, read at a screen of Nodes and at a ring of starts, `docs/designs/light_bending/STEP_ALGEBRA.md` of the archived code [84], on `main` at 23ef9ce9 .
- [45] The algebraic check of flow-link-v1: the step algebra beside a mass, the ring of starts and Newton's rows under the weighted flow, `docs/designs/flow_weight/ALGEBRA.md` of the archived code [84], on `main` at d9b6694a.
- [46] The atom's algebra: two closing radii as a ratio of counts at a detector, and the click that moves a row between them, `docs/designs/atom_algebra/ALGEBRA.md` of the archived code [84], on `main` at 643b7ae7 .
- [47] The ring worlds of flow-link-v1: the pins before the run, the one run of 2026-09-22 and the register's refusal, `examples/events/flow_link/README.md` and `expectations.json` of the archived code [84], on `main` at 23b0e630 (the merge of `flow-link-build`; the run on the split ladder read at the branch head b094c53b, its files unchanged by the merge).
- [48] The dark sector against the law's own rules: no dark energy and no dark matter, does it close the picture, `docs/designs/far_lamp/DARK_SECTOR.md` of the archived code [84], on `main` at aa243019 (a host map, no run).
- [49] The criteria of the confrontation table, row by row, with the re-runs at head, `docs/designs/paper_criteria/CRITERIA.md` of the archived code [84], on `main` at f5fe8004 .
- [50] Lorentz from the clicks: the click frame, its section 0 the click theorem, `docs/designs/click_frame/DERIVATION.md` of the archived code [84], on `main` at 70e9781a (the file identical to its head 1dd81fef).
- [51] The confrontation register, `docs/NATURE.md` of the archived code [84].
- [52] The paper's records, cut from the paper at 48 pages (record 1244 of the log) and kept verbatim from the same source: the full confrontation record, the check of the roads, the seven confirmations, the auxiliary proofs, the hand-worked update, the conversion table and the forcing ledger, `paper/general_formula/records.tex` of the archived code [84].
- [53] J. S. Bell, *Physics* 1, 195 (1964), <https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195>.
- [54] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, *Phys. Rev. Lett.* 23, 880 (1969), <https://doi.org/10.1103/PhysRevLett.23.880>.
- [55] G. C. Ghirardi, A. Rimini and T. Weber, *Lett. Nuovo Cimento* 27, 293 (1980).
- [56] J. Hardy, O. de Pazzis and Y. Pomeau, *Phys. Rev. A* 13, 1949 (1976), <https://doi.org/10.1103/PhysRevA.13.1949>.
- [57] U. Frisch, B. Hasslacher and Y. Pomeau, *Phys. Rev. Lett.* 56, 1505 (1986), <https://doi.org/10.1103/PhysRevLett.56.1505>.
- [58] S. Herrmann, A. Senger, K. Möhle, M. Nagel, E. V. Kovalchuk and A. Peters, Rotating optical cavity experiment testing Lorentz invariance at the 10^{-17} level, *Phys. Rev. D* 80, 105011 (2009), <https://doi.org/10.1103/PhysRevD.80.105011>.
- [59] M. Nagel, S. R. Parker, E. V. Kovalchuk, P. L. Stanwix, J. G. Hartnett, E. N. Ivanov, A. Peters and M. E. Tobar, *Nature Commun.* 6, 8174 (2015), <https://doi.org/10.1038/ncomms9174>.
- [60] J. Bailey et al., *Nature* 268, 301 (1977), <https://doi.org/10.1038/268301a0>.
- [61] B. Botermann et al., *Phys. Rev. Lett.* 113, 120405 (2014), <https://doi.org/10.1103/PhysRevLett.113.120405>.
- [62] U. Sinha, C. Couteau, T. Jennewein, R. Laflamme and G. Weihs, *Science* 329, 418 (2010), <https://doi.org/10.1126/science.1190545>.
- [63] A. M. Gleason, *J. Math. Mech.* 6, 885 (1957), <https://doi.org/10.1512/iumj.1957.6.56050>.
- [64] D. L. Donoho and P. B. Stark, *SIAM J. Appl. Math.* 49, 906 (1989), <https://doi.org/10.1137/0149053>.
- [65] E. H. Kennard, *Z. Phys.* 44, 326 (1927), <https://doi.org/10.1007/BF01391200>.
- [66] H. Weyl, *The Theory of Groups and Quantum Mechanics*, translated by H. P. Robertson (Methuen, London, 1931; Dover, New York, 1950), chapter IV, section 14.
- [67] J. Schwinger, *Proc. Natl. Acad. Sci. USA* 46, 570 (1960), <https://doi.org/10.1073/pnas.46.4.570>.
- [68] H. P. Robertson, *Phys. Rev.* 34, 163 (1929), <https://doi.org/10.1103/PhysRev.34.163>.
- [69] H. Maassen and J. B. M. Uffink, *Phys. Rev. Lett.* 60, 1103 (1988), <https://doi.org/10.1103/PhysRevLett.60.1103>.
- [70] P. Jordan and J. von Neumann, *Ann. Math.* 36, 719 (1935), <https://doi.org/10.2307/1968653>.

- [71] I. Bialynicki-Birula, Phys. Rev. D 49, 6920 (1994), <https://doi.org/10.1103/PhysRevD.49.6920>.
- [72] D. A. Meyer, J. Stat. Phys. 85, 551 (1996), <https://doi.org/10.1007/BF02199356>.
- [73] P. Arrighi, Natural Computing 18, 885 (2019), <https://doi.org/10.1007/s11047-019-09762-6>.
- [74] D. Bohm, Phys. Rev. 85, 166 and 180 (1952), <https://doi.org/10.1103/PhysRev.85.166> and <https://doi.org/10.1103/PhysRev.85.180>.
- [75] D. Dürr, S. Goldstein and N. Zanghì, J. Stat. Phys. 67, 843 (1992), <https://doi.org/10.1007/BF01049004>.
- [76] R. W. Spekkens, Phys. Rev. A 75, 032110 (2007), <https://doi.org/10.1103/PhysRevA.75.032110>.
- [77] N. Harrigan and R. W. Spekkens, Found. Phys. 40, 125 (2010), <https://doi.org/10.1007/s10701-009-9347-0>.
- [78] A. Fine, Phys. Rev. Lett. 48, 291 (1982), <https://doi.org/10.1103/PhysRevLett.48.291>.
- [79] B. S. Cirel'son, Lett. Math. Phys. 4, 93 (1980), <https://doi.org/10.1007/BF00417500>.
- [80] S. Popescu and D. Rohrlich, Found. Phys. 24, 379 (1994), <https://doi.org/10.1007/BF02058098>.
- [81] H. S. Poh, S. K. Joshi, A. Cerè, A. Cabello and C. Kurtsiefer, Phys. Rev. Lett. 115, 180408 (2015), <https://doi.org/10.1103/PhysRevLett.115.180408>.
- [82] B. Hensen et al., Nature 526, 682 (2015), <https://doi.org/10.1038/nature15759>.
- [83] P. Grangier, G. Roger and A. Aspect, Europhys. Lett. 1, 173 (1986), <https://doi.org/10.1209/0295-5075/1/4/004>.
- [84] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), <https://doi.org/10.5281/zenodo.22738746> (the concept DOI, resolving to the latest version).

A Technical details

The tables and their rounding. The circle is $\mathbb{Z}_{N_\phi}$ with $4 \mid N_\phi$. The tables are $C_{N_\phi}[p] = \text{round}(256 \cos(2\pi p/N_\phi))$ and $S_{N_\phi}[p] = C_{N_\phi}[p - N_\phi/4]$, so $C_{N_\phi}[p + N_\phi/2] = -C_{N_\phi}[p]$ and $S_{N_\phi}[p + N_\phi/4] = C_{N_\phi}[p]$ exactly; the norm $C_{N_\phi}^2 + S_{N_\phi}^2$ lies in $[65536 - 351, 65536 + 361]$ for every power of two $N_\phi \leq 65536$ [5]. The normalisation by C_K in Eq. (??) makes the click's probabilities sum to one at completion, and the paper claims exact unitarity nowhere.

The symbols. The symbol table, every letter with its kind and its name, is in the paper's records file [52]; every symbol is named at its first use in the text.

B Reproduction

Every registered run's world file, its expectation written before the run, its source fingerprint and its record, the check scripts with their outputs, the figures' runs and the seven confirmations (each a run's clicks against the formula shown Inside, with its numbers) are listed in the paper's records file [52] and in the registers [4, 7]; the code and the worlds are the archive [84]. The archived version cited at submission is the tag `paper-2026-09-22` on the merge commit of the pull request that carries this cut, one tree whose history holds the trees named above and the log records this paper cites; it replaces the concept DOI of the data availability statement. The software, the computations, the derivations' checks and the text were made with a large language model as a tool under the author's direction and instructions, as the declaration at the end of the paper states; every decision is the author's. Every theorem, identity and number of the paper was re-derived and recomputed independently, one line per statement [17]. The paper's law is the massive record kind with the local detector law, on `main` at `a867ab22` [9]; the beam law's runtime before the generic entry of the bending (the runs' own commits named above, on the first-parent line of `main` up to `3a9a7109`, the last commit before the entry's merge `5b855791`) is history, its flight table and its readings in the records file [52]; the generic entry is a later rule of that runtime (Section 6.1, P9), and the ring's run (the full record [51, 52], row 13) is this paper's one reading under it. Every record, note, design and register row this text cites is on the tree of this cut (main merged into it at `87c7ec6b`).

The long form of this manuscript, with the derivation record of every formula, is the tree's paper at the commit `c15c1174`; the forty-page cut before it at `b0d1ebf7`.